Technological Development and Industry Trends of Unmanned Aerial Vehicles (UAVs)

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ABSTRACT

The Taiwan Testing and Certification Center (ETC) supports Taiwan's government by promoting product safety and consumer protection through testing and certification services. As Taiwan develops its UAV industry, the ETC aims to build a standardized UAV testing and certification framework aligned with international standards to strengthen Taiwan's position in the global market. This project developed a high-level research and implementation plan for UAV testing by examining global UAV regulations, requirements, and market entry practices. We consulted with testing and certification experts to understand the current processes and Taiwan's position in the industry. We created an interactive interface to compare international regulations and connect them to Taiwan's testing practices.

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We also are thankful to the drone manufacturers AVIX and Carbon-Based Technologies Inc., who let us meet with them and tour their facilities. They also provided feedback on our final product as end users.

We would also like to thank our advisors, Professor Hung and Professor Snoddy, for their continuous support, feedback, and direction. Their guidance helped shape our project and improve the quality of work.

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EXECUTIVE SUMMARY

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are becoming increasingly important in many industries including defense, transportation, inspection, and agriculture. As a result, countries around the world are investing in this new technology, including Taiwan. Motivated by national security concerns and economic growth, the Taiwanese government is heavily investing in domestic development of UAVs and subsidizing companies working towards growing the UAV industry. Just like any other industrial objective, this goal will not be easily attainable. Fortunately, Taiwan is no stranger to being a global technology leader and operating in the high-tech sector with control of over 60% of the global semiconductor market. Based on Taiwan's track record, it possesses both the technical capability and manufacturing foundation to become a major producer of UAVs for the global market.

In the current UAV global climate, China holds the most control and expertise, with a company called DJI based in Shenzhen, China, controlling about 80–90% of the current UAV consumer market. Due to the growing importance of drones, especially in military contexts, major consumer regions like the United States and European Union are looking for a reliable supplier outside of China, or any other "red" country. This shift toward a non-red supply chain, not to mention Taiwan's unique relationship with China, and an increased global demand for drones, provides Taiwan with the opportunity to expand its stake in the non-red UAV supply chain to the benefit of the international and domestic markets. Taking advantage of this opportunity, however, requires more than just strong technology; it also requires the ability to meet shifting international standards. There is no unified global system for testing and certifying UAVs across different markets. Each country has its own rules and regulations. The rules are also constantly changing to keep up with technological advances. Taiwan's UAV manufacturers do not have a clear path to drone certification for the global market due to the lack of internationally accepted regulations.

A key example of this challenge can be seen in the United States. In December 2025, an executive order by the Trump administration banned all new, foreign-made UAV models without a certain "Blue UAS" certification. These standards focus on device security and supply chain control (building toward a 'non-red' supply chain). Companies that do not meet these requirements cannot compete in the American market. The European Union also controls how UAVs are imported. In the EU, there are three categories based on risk—open, specific, and certified. Each has different rules for pilots and operators, where higher risk operations require greater levels of approval and oversight. This means that certification is not only about the product but also how it is used. As a result, companies must design their UAVs to meet standards and plan how their UAVs will be used. This process takes time and adds to the cost of developing and selling UAV systems. Without clear guidance, companies may fall behind or avoid entering global markets altogether.

Based on these challenges, the project sponsor, Taiwan Testing and Certification Center (ETC), faces challenges in providing quality testing and certification services to the emerging

UAV market. Addressing this challenge will help the ETC better support manufacturers that want to enter global markets. Taiwan's position in the international UAV industry will improve as it increases production domestically and internationally.

To address this problem, a team of four WPI students worked with the ETC and traveled to Taiwan for their IQP. The team includes two robotics engineering students, one mechanical engineering student, and one information systems and technology student. The goal of the project was to support the ETC in aligning Taiwan's UAV certification system with international standards. To achieve this goal, the team studied the EU's, U.S.'s, and Taiwan's regulations. This allowed the team to understand what requirements manufacturers must meet and compare them across the three markets as they create recommendations and tools to support ETC services.

To produce quality recommendations and tools for the ETC at the end of a 7-week period, the team worked with ETC staff to learn about their current testing procedures, through a combination of staff interviews and facility tours. Tours included UAV focused areas: electromagnetic compatibility, safety, reliability, and cybersecurity testing. Through correspondence and materials shared with the team by the ETC, the team was able to gain an understanding of the ETC's technical and business capabilities, revealing the gaps that exist within the current regulatory system the ETC follows in their developing UAV testing service.

In addition to the materials from ETC, the team visited two drone manufacturers, AVIX and Carbon-Based Technologies, with help from the ETC. These two companies mostly operate in Taiwanese markets and are exploring opportunities to enter the international scene. During our visits, the team discussed the challenges related to export certification, international regulations, and presented findings on United States Blue UAS requirements. These discussions demonstrated that there is a major barrier for Taiwanese manufacturers. Even when companies have strong products, they may not be able to enter new markets due to unclear or difficult certification requirements. This further emphasizes the importance of the project.

After the team gathered materials regarding regulations, frameworks, and standards, the information was organized for analysis. The team defined three main objectives for the project's outcome. These objectives support the ETC's mission of providing UAV testing and certification services that ensure compliance with both domestic and international standards. These three objectives are:

- Comparative analysis of ETC and online-sourced materials
- Discovering a pathway for entering the U.S. and EU markets
- Developing a UAV export and regulation interface that the ETC can use to navigate different UAV regulatory frameworks.

The team's analysis consisted of a comparison of specific standards and qualifications between the ETC and U.S.–EU regulatory systems as well as a thematic comparison. This comparison demonstrated a large gap between the ETC's capabilities and international requirements. Only a small portion of the standards used in the U.S. and EU were found in the ETC's current system, with an overlap of approximately 3.3% similarity of standards. However, 75% of the topics in which the ETC expressed interest showed thematic similarities to both U.S.

and EU regulations. Although limited by the number of materials from online and ETC sources, as well as the team's standard extraction method in capturing transferable standards between the ETC and U.S.–EU market, this analysis showed that the ETC still has a long way to go in aligning with regulatory systems abroad. Fortunately, with the ETC having 75% similarity in UAV-related topics, future regulatory alignment in this emerging industry is possible.

To build on the findings from our analysis, the team conducted research that revealed pathways to enter both the U.S. and EU markets. There are two levels of certification for entering the U.S. market, Blue UAS and Green UAS certification. Blue UAS certification primarily focuses on defense utilization and is a more onerous application process. Although there are ways that companies can bypass some of the red tape, Blue UAS certification is still expensive and complicated, especially for foreign companies. Our research team recommends that a more streamlined solution for the ETC is to attain Green UAV certified assessor status through the Association for Uncrewed Vehicle Systems International (AUVSI). AUVSI created the Green UAS program, which is focused on commercial and non-defense UAV certification. It is also an entry point into future Blue UAS certification. AUVSI endorses and then contracts with testing bodies to perform the actual UAV certifications tests. Another Taiwanese institution, Industrial Technology Research Institution (ITRI), has already become a certified assessor for Green UAS cybersecurity testing, showing that it is possible for an institution based in Taiwan to get this certification. For these reasons, it is our recommendation that the ETC apply to become a Green UAS certifying institution. Receiving Green UAV certification status would validate the tests the ETC performs which are required for non-defense UAVs to be sold in the United States. We have provided the ETC with a list of essential contacts in the United States, including AUVSI. Because of our project, ETC representatives will be visiting AUVSI's headquarters in Arlington, Virginia, in July 2026. Overall, these steps position ETC to expand its role in UAV certification and support Taiwanese manufacturers entering the U.S. market.

Entering the EU market requires manufacturers to work with an authorized representative, defined as a legal entity within the EU that acts on behalf of the manufacturer. This representative supports the conformity assessment process, which must be approved by the National Aviation Authority (NAA) and leads to a Declaration of Conformity, the key requirement for selling UAVs in the EU market. There is no alternative certification pathway to EU regulation 2019/945 or comparable scenario such as how Green UAS certification provides an entrée to Blue UAS certification in the U.S. By following EU regulation 2019/945 and providing a conformity assessment service, with an accompanying Declaration of Conformity, the ETC can support Taiwanese UAV manufacturers access to the EU market.

To achieve our last objective—sustainably supporting the teams' recommendations—we have developed a UAV regulation website that organizes UAV certification information. Currently, the website brings together regulations from Taiwan, the U.S., and the EU. The team gathered this information from a variety of sources including the ETC and online sources. The website can be updated by the user, and the team strongly recommends that ETC keeps it up to date as new developments frequently occur in the regulatory environment. To navigate the website, information is structured by region, topic, and certification category. This makes it easier to gain knowledge across different markets. The website reduces regulation-related confusion by consolidating essential information in an easily accessible format; ETC staff can also update and expand the website's datasets over time. This tool is designed to support both ETC staff and external stakeholders. Manufacturers can use the website to understand certification requirements before attempting to enter new markets. We recommend that the ETC send this website to manufacturers interested in learning about the import regulations of various markets. Overall, this regulatory website serves as a practical, scalable tool that simplifies access to certification information. It supports both the ETC and UAV manufacturers in navigating international requirements efficiently.

With all 3 objectives achieved, the team was not only able to provide recommendations to the ETC on navigating international regulatory frameworks and applying for Green UAS testing certification/ fulfilling European Union Aviation Safety Agency (EASA) requirements but also a tool to carry out these recommendations. In conclusion, this project addressed a key problem in Taiwan's UAV industry: the lack of a clearly defined certification pathway for entering international markets. While Taiwan has strong technical capabilities, that alone is not enough to overcome certification barriers. As a result, the ETC was faced with the challenge of providing the required testing services for UAV certification to their domestic and international customers in the UAV industry. This project intends to help the ETC align its current and future UAV testing services with global standards. To complete this work, a team of four WPI students used several research methods consisting of a literature review, contextual analysis, comparative and thematic analysis, to develop solutions and recommendations for the ETC. These recommendations include ways of navigating international regulatory frameworks, applying for Green UAS testing certification/ fulfilling EASA requirements and the utilization of a UAV regulation interface developed with a baseline of the U.S.'s, EU's, and Taiwan's frameworks. By giving the ETC these recommendations and tools, it will provide the UAV industry in Taiwan with a clearer pathway to achieve international certification and accelerate the expansion of Taiwan's UAV capabilities on an international scale.

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Unmanned Aerial Vehicles (UAVs), commonly known as drones, have been established in the world market across many industries. Now, the Taiwanese government is placing greater emphasis on developing its asymmetric warfare capabilities, reducing dependence on Chinese technology, and moving away from “red supply lines” to improve national security (Danielsson, L., 2026). Additionally, the diverse set of significant roles these technological devices play across a multitude of applications—national security, human resources, and consumer entertainment—makes it an important development for today’s society.

Alongside strengthening the domestic UAV market, Taiwan has goals to expand globally, to become a supplier and manufacturing collaborator in the international drone market. As a result, Taiwan is looking to beef up its technological infrastructure to support development in this area, but it must also establish sound regulations that will guide the changes and development of production. To achieve this, Taiwan must not only invest in technical but also market research on this technology.

Although Taiwan is at a good technical position to become a strong, trusted supplier of UAVs, especially with the island’s reputation of controlling 72% of the global chip market (Nasdaq, 2026), the lack of robust UAV testing practices, certification, and regulation is holding back practical implementation, manufacturing, and market strategy for Taiwan’s UAV industry. Coupled with differences in international standards and a lack of global guidelines, expansion into the global market will require a regulatory alignment between Taiwan and international bodies.

Since October 2019, there have been tremendous developments in Taiwan that enable the operation and use of UAVs through organizations such as the Civil Aviation Administration (CAA), the National Communications Commission (NCC), and the Bureau of Standards, Metrology, and Inspection (BSMI). However, these developments are not yet sufficiently advanced to provide UAV manufacturers (makers) in Taiwan with proper guidelines that enable the mass development of UAVs for different operational applications, or to be well aligned with international standards, such as those used in the United States and European Union, for easy trading in the global market.

The sponsor, a government-affiliated testing and certification company (ETC), tasked this research team for a comprehensive list of international UAV regulations, ranging from reliability and component testing to international standards and regulation alignment, supporting Taiwan’s growing service demand in UAV testing and certification. The main objective of this project is to create a detailed plan to align Taiwan’s UAV regulations and certifications with global requirements, and position Taiwan to become an industry leader in the development and manufacture of UAVs. To fulfill these objectives, extensive literature reviews and in-depth research were conducted on a variety of different technologies necessary for a comprehensive regulation alignment plan.

For our technology and regulation research, this paper covers literature research into the following: Past and current United States and European Union drone regulations and requirements, current Taiwanese certifications and regulations for drone development, market export strategies for countries entering a new market, and a variety of reliability testing practices, ranging from environmental sustainability, electromagnetic capabilities, radio frequency communication, and other important UAV technologies.

This paper was divided into four sections: background and literature review, research methodologies, results and analyses, conclusions, and recommendations. The background and literature review section went into the regulatory aspects of drone development. The research methodology section outlined our research components, data collection methods, and deliverable developments. These included questionnaire responses from ETC's subject-matter experts, recommendation analysis, and website development. The results and analysis sections detailed the findings and final deliverables of this project, including recommendations to help the ETC continue its path toward becoming a certification body for Taiwan's ever-developing UAV industry, while the conclusions and recommendations chapters synthesized the research report.

Chapter 2: Background

2.1: Unmanned Aerial Vehicles (UAV)

An unmanned aerial vehicle (UAV) is an aircraft that is controlled remotely or operates autonomously. UAVs do not have a human pilot, crew, or passengers on board. Although they come in many shapes and sizes, they are often referred to as unmanned aerial systems (UAS), remotely piloted aircraft systems (RPAS), or more commonly, drones (Altigator, n.d.). Inside these vehicles are integrated systems that include motors, propellers, batteries, cameras, and sensors (BeyondSky, 2026). Each component has its own function and limitations, but together they form a cutting-edge system. UAVs can take different forms such as quadcopters, hexacopters, fixed-wing aircraft, or single-rotor designs, each suited to specific operations (Pigeonis, n.d.). They provide diverse benefits in transportation, mobility, and environmental monitoring. For example, in a study on comparing behaviors of photographers using unmanned aerial vehicle in China and Europe, it was found that UAV photographers have expanded their photographic perspective into three-dimensional space compared to traditional photography due to the UAV capabilities (Chen et al., 2023). This is also seen in research conducted by Missouri EPSCoR researchers. The research investigated the accuracy of low-cost sensor fusion used for estimating crop biophysical and biochemical parameters, which was only accessible due to UAV information fusion capabilities (Maimaitijiang et al., 2017). This demonstrated that UAV's serve as powerful tools for information collection and analysis, which could be used in decision making applications (NESAC, n.d.; EBSCO, 2024). Lastly, UAVs are redefining defense strategies by offering remote operability and inexpensive manufacturing which enables scalable deployment for military operations. For example, UAVs have played a significant role in the Russia–Ukraine war, where drones helped level the battlefield by enabling Ukraine to counter larger Russian forces and save lives with their high-level UAV production (CBS News, 2025). As this technology develops, proper regulations are necessary to ensure safety, management, and effective commercial and military integration.

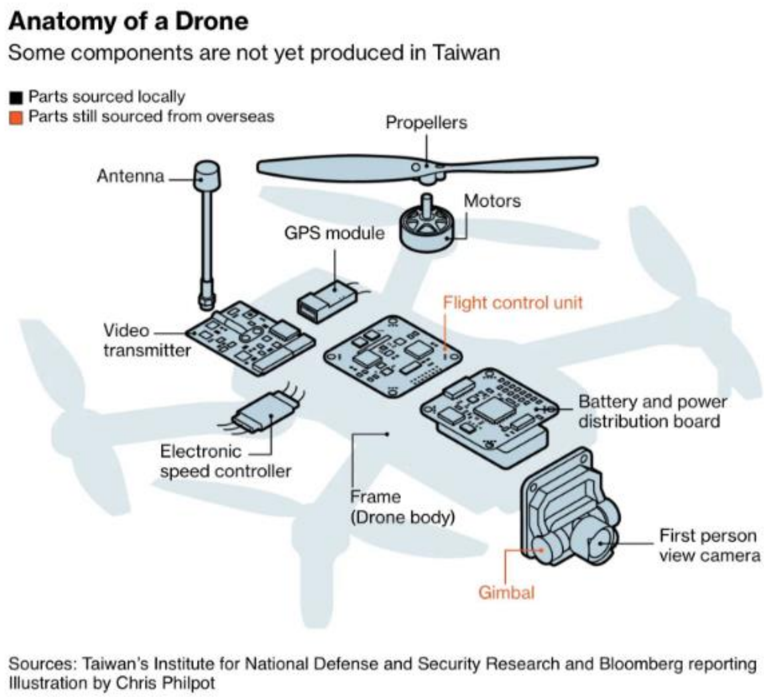

Figure 1: Visual diagram that shows all the parts of the UAV (2025)

A major factor behind UAVs' ability to perform multifunctional operations lies in their core components. While UAV systems may include various sensors and network modules, there are fundamental components common to all UAVs. These include propellers, motors, power systems, electronic speed controllers, flight controllers, and payload systems (Um, J. S., 2019). Together, these parts enable UAVs to achieve their primary function, controlled flight.

2.1.1: Flight Controllers

For a UAV to stay stable while in flight and respond correctly to controller inputs, there must be a central computer onboard that processes flight information and makes control decisions. This component is called the flight controller. While the electronic speed controller manages the electrical power delivered to each motor, the flight controller acts as the brain, taking data from different onboard sensors and using it to determine how the UAV should move. In most UAVs, the flight controller works as a part of a larger flight control system that includes onboard processors, estimation software, and communication links (Kangunde et al., 2021). Most modern flight controllers rely on an inertial measurement unit (IMU), which includes accelerometers, gyroscopes, magnetometers, barometers, and sometimes GPS, depending on the required level of control and automation (PX4, n.d; GuideNav, 2024). By combining information from the sensors, the flight controller can estimate the UAV's position, direction, altitude, and orientation, allowing for in-flight self-correction to maintain a stable path, even when affected by wind or other outside forces (Kagunde et al., 2021). As a result, the flight controller is one of the most important components in the UAV system; it coordinates sensors, flight logic, and actual aircraft movement into one process.

2.1.2: UAV Motors

UAV motors convert energy generated from an electrical power source (such as a battery) into mechanical motion. This is similar to car engines which generate mechanical motion through combustion of fuel; The powerful rotational motion produced by the drone motor is used to produce thrust by the drone's propellers. This effectively transforms electrical energy into linear motion. Because each UAV can have multiple motors that can spin at different speeds, the drone is able to move with several degrees of freedom.

The specific common motors in UAVs are brushless DC (BLDC) motors (D. L. Gabriel, J. Meyer and F. du Plessis, 2011) composed of three main components, a rotor, stator, and controller (Stanley, M. 2025). The stator (coil windings) is placed inside the rotor (a circular permanent magnet). When direct current (DC) power is applied to the coils, the winding coils turn the stator to an electromagnet. The interaction between the electromagnet and the permanent magnet in this configuration causes the rotor to spin continuously while the battery supplies DC power. To maintain smooth rotation of the rotor around the stator, the controller (computer chip) calculates the rotor's position and switches the polarity of the stator as needed, ensuring continuous motion (Circuit Bread, 2026). Apart from ensuring the continuous movement of the rotor, the controller (also known as an electronic speed controller) performs other functions (mechtex, 2026).

2.1.3: Electronic Speed Controllers

Electronic speed controllers (ESC) manage the speed and direction of the UAV during flight, compensating for wind and other external forces. With the help of the computer/chip, the ESC regulates the electrical charge delivered to each coil, enabling precise control of motor speed and orientation.

Since the main source of power to this controller is Direct Current (DC), ESC uses its metal oxide semiconductor field effect transistor (MOSFETs) to switch the DC into alternating current (AC) (Aventura All Remote & Smart Life Solutions 2022). This AC current then keeps the rotor spinning smoothly due to its changing magnetic field.

Additionally, as the rotor magnet passes the coils, voltage is induced. The ESC can detect this voltage using a comparator and an analog-to-digital converter (ADC) (FPV University and Hacker University, 2021). This feedback helps the ESC determine rotor position, predict future movement, and adjust coil energization, preventing motor damage, and ensuring efficient operation.

2.1.4: Propellers

While motors generate rotational force, UAVs require propellers to convert that force into thrust for lift and maneuverability. A propeller consists of angled blades mounted on a hub, which connects to the motor's output shaft. As the blades rotate, they push against the air, generating thrust (Kanika Madaan, 2026). Though simple in design, propellers are critical to drone performance. Parameters such as diameter, pitch, blade count, and material significantly affect flight characteristics, including speed, thrust consistency, and load capacity (Ligpower, 2025).

2.1.5: Power Systems and Batteries

In order to operate, all UAVs must have a reliable source of power, i.e., either a battery or liquid fuel. A UAV's power system generally includes the battery, power distribution components, voltage regulation hardware, and monitoring systems that deliver electricity to the rest of the drone (PX4, n.d.). Many recently manufactured UAVs use lithium-ion and lithium-polymer batteries because they have a good weight to energy storage ratio, which is important for aircraft that need to stay light while having a useful flight time. However, battery systems also create performance limitations, since endurance, payload capacity, and range are directly affected by how much energy the battery can store and use efficiently (Townsend et al., 2020). UAV battery systems frequently rely on monitoring functions that track voltage, current, temperature, and remaining charge so that the aircraft can avoid overuse, overheating, or power loss during flight (PX4, n.d.; Zhao et al., 2025).

2.1.6: UAV Payload Systems

While motors, propellers, batteries, and controllers are needed for a UAV to fly, payload systems are components that allow the UAV to perform a specific task. Put simply, the payload is the specific equipment carried by the aircraft and determines the actual function of the UAV for applications such as mapping, agriculture, transport, or consumer use (Kovanič et al., 2023; Xu et al., 2021). This may include cameras, infrared sensors, light detection and ranging (LiDAR), spraying systems, or delivery mechanisms (YAO et al., 2025). Payload systems can also affect UAV design, as different payloads create different weight, power, mounting, and data processing requirements (Yao et al., 2025). A lighter payload may allow the UAV to fly longer and move more efficiently, while a heavier or more complex one may reduce endurance and require stronger propulsion and control systems. Because of this, payload systems are not just extra attachments, but vital parts to UAV design that influence performance, mission capability, and aircraft requirements.

2.2: UAV Testing and Certification

Like any other complex technological system, proper testing and regulations are necessary to ensure safety, ethics, and reliability. For example, a comparative analysis published by Canadian Science Publishing highlighted that UAVs have become safer as certification, regulation, and safety-related technologies are developed (Devon Wanner et al., 2024).Without the regulations that guides developing UAV technology, the design of these new technologies include specifications that might bring more harm to people or the environment.

Regulations apply to UAV operators, pilots, and manufacturers, and are established by regulatory bodies with authority over the airspace in which UAVs operate, such as the FAA, or in which they are sold, such as the European Union. With growing interest, significant technical involvement, and the potential for industry disruption, both public and private organizations are advancing UAV regulation and certification. These efforts range from incident reporting and manufacturing standards to pilot licensing (Sándor & Pusztai, 2021).

Regulatory frameworks and guidelines across regions vary, most evident in the three regions of interest this research project addresses: the United States, the European Union, and Taiwan. The Taiwan Testing and Certification Center serves as an assessor for many of Taiwan's certification requirements, including, but not limited to, UAV systems.

2.2.1: Taiwan Testing and Certification Center

The Taiwan Testing and Certification Center (ETC), established in 1983, is a government-designated, non-profit organization that provides product testing, inspection, and certification services to ensure compliance with domestic and international standards. Formerly known as the Electronics Testing Center, the organization was renamed in 2020 to move towards diversified development and expand its service scope.

The ETC operates under the supervision of Taiwan's Ministry of Economic Affairs (MOEA) and plays a strategic role in supporting Taiwan's industrial development and export competitiveness. As a government-supported entity, ETC helps manufacturers verify that products meet regulatory, safety, performance, and quality requirements before entering domestic and international markets. Its services span multiple high-technology sectors, including electronics, communications equipment, aerospace components, and emerging unmanned aerial vehicle (UAV) systems (Taiwan Testing and Certification Center, 2023).

Within the UAV industry, the ETC contributes to Taiwan's broader national strategy of strengthening "trusted industries" by providing technical validation, conformity assessment, and certification support (Office of the President Republic of China (Taiwan), 2024). By ensuring compliance with evolving regulatory standards and international market requirements, the ETC supports supply-chain security, product reliability, and global market access for Taiwanese manufacturers.

In this project, ETC serves as both the sponsoring organization and a central stakeholder in Taiwan's UAV certification ecosystem. Its institutional capacity, technical expertise, and regulatory alignment are critical factors in advancing efficient, scalable, and internationally competitive UAV testing and certification pathways.

2.2.2: United States UAV Development

UAV development and regulation is advancing rapidly in the United States and the European Union. In the U.S., the Federal Aviation Administration (FAA) regulates civil aviation, overseeing aerial activities nationwide, while the Federal Communications Commission (FCC) regulates interstate and international communications, including cable, radio, and satellite. Together, the FAA and FCC shape the regulatory landscape for UAV operations.

Recently, the United States has been accelerating the integration of UAVs into the National Airspace System, to remain a competitive player in the global UAV industry, where China is currently the world leader, holding almost 90% of the market. By working more closely with domestic manufacturers, the U.S. hopes to continue developing its UAV industry. To support this effort, the White House issued the executive order "Unleashing American Drone Dominance" to upscale commercialization of U.S. drones and strengthen the domestic industrial base (Greenberg Traurig LLP, 2025). This policy is not simply about increasing drone usage in the United States. It is also about reshaping the U.S. drone industry by reducing reliance on foreign made systems, and creating stronger domestic and trusted supply chains. In this understanding, the executive order is both for commercialization and industrialization. It aims to expand UAV operations in the National Airspace System and has important implications for Taiwan.

As the United States moves away from other suppliers, it will require new manufacturing and supply partners that can meet its requirements for quality, cybersecurity, and other regulations. Taiwan is well positioned to benefit from this change because of its strong manufacturing base and growth potential. Partnership also makes certification alignment more practical. If Taiwan can align its framework and certification standards to meet the U.S. standards, manufacturers in Taiwan can demonstrate systems compatibility. This could lead to reductions in entry barriers and make Taiwan a more attractive source of system and component procurement.

In addition, the U.S. Department of Defense (DoD) launched the Replicator Initiative, which aims to field thousands of low-cost, attainable drones. These efforts are reflected in the work of multiple companies, including new startups engaged in different stages of UAV production for diverse applications in the U.S. market. The Center for Security and Emerging Technology (CSET) conducted an analysis of the Association for Uncrewed Vehicle Systems International (AUVSI) USRD database and published a study in November 2025 that identified approximately 2,506 UAVs available in the U.S. market for top 10 applications (Miller, Bresnick, Feldgoise, & Schoeberl, 2025). Meanwhile, the Federal Aviation Administration (FAA) reported 377,000 recreational UAV units and 433,000 commercial UAVs registered in the U.S. as of November 2025 (Federal Aviation Administration, 2025).

Although, 2,506 is not the exact number of UAV models manufactured in the U.S. or operating domestically as the number includes procured drones and omits drones used in less popular applications, it provides a snapshot of the progress U.S. manufacturers are making toward the objectives outlined in both the executive order and the DoD's initiatives. With growing innovation and production capacity, the U.S. drone industry is poised for further expansion, particularly as it seeks to build a secure, trusted supply chain and establish a stronger presence in the global technological landscape.

2.2.3: Blue and Green UAS Certifications

Blue UAS certification is a Department of Defense program, created in 2020 and previously managed by the Defense Innovation Unit (DIU), that oversees drone research and development. The program was created to address security concerns, supply chain control, and compliance with the National Defense Authorization Act (NDAA) (Defense Innovation Unit, 2025a). Drone manufacturers must submit their systems for review before they can be considered Blue UAS certified.

In the original version of Blue UAS certification, manufacturers often needed government involvement. Drones were approved through sponsorship by a branch of the Department of Defense, such as the U.S. Air Force or Army. This early version of Blue UAS certification limited the number and kind of manufacturers that could enter the system to defense contractors only.

This process began to change in May 2025, when the DIU announced updates to expand the program by establishing a group of third-party-recognized assessors who reviewed drone platforms, components, and software, and then sent reports to the government for final approval (Defense Innovation Unit, 2025b).

Subsequently, in July 2025, DIU recognized Green UAS certification as a pathway to Blue certification. Green UAS certification was created by the Association of Uncrewed Vehicle Systems International (AUVSI). This was possible because AUVSI is a recognized third-party assessor for Blue UAS. The green UAS certification program allows manufacturers to meet the same standards without direct government sponsorship (Defense Innovation Unit, 2025c) (AUVSI, 2025a), and follows the exact same standards as Blue UAS, but uses a simpler certification process (AUVSI, 2025a). AUVSI does not conduct testing itself and instead delegates them to a third-party assessor, similar to Blue UAS.

To obtain certification, manufacturers apply and submit detailed forms for the platform's, components and software. AUVSI then matches manufacturers with a third-party assessor who reviews the system and produces a report for AUVSI, who can then choose to approve or deny the application. After completing Green UAS, manufacturers can move into Blue UAS, where the same materials and reports are submitted for final review, with the help of AUVSI. If approved, the system is added to the Blue UAS list (Defense Innovation Unit, 2025b).

Both Blue and Green UAS certifications require the same tests, which include cybersecurity audits, supply chain checks, penetration testing, and data security reviews. They also include hardware and software validation, as well as documentation review, for manufacturers seeking certification (Defense Innovation Unit, 2025b).

The Green UAS program relies on qualified third-party assessors who must be vetted and authorized by AUVSI. The process of being recognized as a Green UAS certifier by AUVSI is much simpler than being recognized as a Blue UAS certifier. Taiwan is making headway to obtain this certification. In January 2026, Taiwan's Industrial Technology Research Institute (ITRI) was accepted as a recognized cybersecurity assessor in the Green UAS program. This enables them to conduct the software penetration testing required for Green UAS (AUVSI, 2026). They were the first and remain the only non-American recognized assessor for the Green UAS program.

The Green UAS certification program has two levels: Green UAS Cleared is meant to follow the same regulations as Blue UAS certification. Because of this, following the Green UAS Cleared process is the most direct pathway to attain the more stringent Blue UAS certification. Green UAS Certified is more detailed. In addition to the same requirements that are found under Green Cleared, there are additional requirements, including an assessment of the entire company and its internal systems. Green UAS Certified is not required to get on the Blue UAS cleared list, but it shows another level of security that some consumers and especially governmental, entities may require (AUVSI, 2025b).

In December 2025, new rules increased the importance of obtaining Blue UAS certification; the FCC now requires new foreign-made drones to receive Blue UAS certification before they can be imported. This rule primarily targets Chinese manufacturers and limits new models from entering the U.S. market, although drones that already had approval could still be imported. Companies that build the final drone platform in the U.S. and use at least 65% domestic components are not required to obtain this certification (Federal Communications Commission, 2025). Manufacturers may also obtain special clearance from the Department of Defense rather than through the Blue UAS program; this, however, is typically harder to obtain and applies only to military applications.

Also in December 2025, DIU announced that management of the Blue UAS program would transition to the Defense Contract Management Agency (DCMA). This change shifts responsibility for maintaining and expanding the program from DIU to DCMA. The transition is intended to scale the program and improve the Department of Defense's management of drone procurement (Defense Innovation Unit, 2025d).

2.2.4: European Union

The European Union Aviation Safety Agency (EASA) develops and enforces aviation safety regulations across European Union (EU) member states. Working closely with the EU governing body and national civil aviation authorities (CAAs), EASA provides certification, regulation, and standardization services to ensure civil aviation safety.

Because the EU is made up of 28 member states and the four European Free Trade Association (EFTA) countries (Switzerland, Liechtenstein, Norway, and Iceland) that participate in the EASA system due to ties with the European Union, more than one organization is involved in shaping UAV regulations that now exist in the EU. These organizations include European Organization for Civil Aviation Authorities (NAAs), European Organization for Civil Aviation Equipment (EUROCAE), EUROCONTROL and Joint Authorities for Rulemaking on Unmanned Systems (JARUS). The organization's functions are as follows:

- The NAAs exist in each European member state and are the agencies responsible for implementing and enforcing aviation regulations within their respective jurisdictions. They work with EASA to ensure compliance with European standards and regulations.
- The EUROCAE develops standards for aviation equipment and systems, providing guidance and technical specifications for the design, manufacture, and operation of UAVs.
- EUROCONTROL is a pan-European organization responsible for coordinating Air Traffic Management (ATM) across European airspace. They work with EASA and NAAs to integrate UAVs into existing ATM system safety and efficiently.

- JARUS is a group of international experts that works with EASA and EUROCTROL to develop harmonized technical, safety, operational standards and guidelines for the regulation of UAVs, ensuring uniform integration of drones into national airspace systems and helping avoid duplicate efforts in the international scene.

Other agencies that support this mission are SESAR Joint Undertaking (SJU), the partnership which provides research and innovative services to modernize and harmonize ATM in Europe, and; the European Defense Agency (EDA) which works with the aforementioned agencies to ensure coordination and compatibility between military and civilian drone operations to promote safety and security—pillars supporting UAV regulations in the EU (EASA, 2024).

In 2018, the set of common rules for civil aviation known as Regulation EU 2018/11139 (aka EASA Basic Regulation) was established. This included a comprehensive framework for the management of all UAV devices, defining governmental or public service functions in the EU, establishing safety and risk-based regulations, and clearly delineated responsibility between EASA and member states (EASA, 2024). After this regulation came the new UAS Regulatory framework that came into effect in 2019, which established regulations for UAVs sold on the EU market. The framework is comprised of two legislative acts, Regulation EU 2019/945 and Regulation 2019/947. These acts continue to define how UAVs can be developed and operated across all EU member states to the present day.

Today, these two legal frameworks define how UAVs can be developed and operated across all EU member states. Regulation 2019/945 focuses on technical certification requirements and third-country certification. This was the first regulation entered into force, with full implementation at the beginning of 2024. Under this act, there is a particular focus on class identification and product compliance of UAV devices. In practice, it defines how manufacturers can design their UAVs using UA classes and applicable technical and compliance procedures to meet EU market compliance requirements. As part of this regulation, Conformité Europééne (CE) marks, like the one shown in Figure 2, were established to classify UAVs sold in the European Economic Area (EEA), which are to be made visible on a UAV device. The CE mark serves as verification that the UAV meets the health, safety, and environmental standards required for marketing throughout the EU, essentially acting as the manufacturer's declaration that their product is in compliance with essential requirements of EU directives. Without the CE mark, there is no alternative method for a UAV to be sold in the EU.

Figure 2: A photo of EU required UAS CE Marking (UAS Certification 2023)

Currently, there are six types of CE marks (C1—C6), each indicating the UAV's technical and safety characteristics, such as weight, speed, noise, remote ID, and geo-awareness capabilities (Class Identification Label (CIL), 2024). Other requirements set out in the regulation document, in tandem with the CE mark determination and affixation, include the development of technical documentation, performing conformity assessments using modules outlined in the regulation document, issuance of an EU declaration of conformity for each product type, and archiving EU declaration documents for 10 years after the product has been placed on the market. Lastly, a reliability requirement was also established for noise emissions following the EN ISO 3744 standard (Regulation 2019/945, 2019).

Although not directly pertaining to manufacturers, the second act, Regulation 2019/947, establishes a risk-based framework that is the very basis of the EU's UAV system. This regulation provides detailed provisions for the operation of UAVs and personnel, including remote pilots and organizations involved in those operations (SKYbrary Aviation Safety, 2019). Under this legislation, three categories of UAS operations are defined: Open, Specific, and Certified. Each of these categories is based on the risk level of the operations it defines.

Specifically, the Open category applies to all UAVs with:

- A total weight of less than 25kg
- Visual line of sight (VLOS) operation
- Maintaining a safe distance
- Remaining below a maximum height of 120 m above ground
- Prohibition from carrying dangerous goods or dropping any materials
- Having a CE mark

The reason for a CE mark requirement in this category is that open categories are divided into subcategories (A1—A3), which further classifies operations using CE mark identification labels, as shown in Figure 3. Subcategory A1 defines whether the flight is over people; A2 defines whether the flight operation is close to people, and A3 is the subcategory for flying far from people. The CE mark informs the type of operator training and whether the drone is allowed to perform certain flight activities. As a result, pilots need to take their UAVs' CE mark into account when planning their flight operations.

The specified category is for operations with a medium level of risk that cannot be performed in the open category. Increased risks include:

- BVLOS (Beyond Visual Line of Sight) flights
- Operations that are over 120 meters high
- Drones over 25kg
- Urban flights with drones over 4 kg or with Electric Conspicuity (EC) certification
- Material dumping
- Flight over crowds of people

As a result, pilots need to meet additional requirements and obtain certification. For example, registration of the UAV operator is necessary and operators must conduct a risk assessment either through a predefined risk assessment (PDRA), specific operations risk assessment (SORA), submission of a declaration to relevant NAA if planned operation fits one of the two published standard scenario (STS-01 or STS-02) or to self-authorize if the operator holds a Light UAS Operator Certificate (LUC) with appropriate privileges (Espila, 2021 and Beyond SKY, 2026)

The certified category defines the high-risk operation that mimics traditional aviation. These operations include:

- A UAV that is designed for transporting people
- Has a characteristic dimension of 3 m or more
- Designed to be operated over assemblies of people
- Designed for transporting dangerous goods
- Hauling cargo and flight in complex environments.

Because of the nature of these UAV operating conditions, no risk assessment performed in the specified category can adequately mitigate the high-level of associated risk without certification of the UAV and the UAV's operator with remote pilot licensing. Additionally, this category is subject to applicable operational requirements in regulations 932/2012, 965/2012, and 1332/2011 (Regulation 2019/947, 2019 and Beyond SKY, 2026). In sum, this category involves a more rigorous certification process for UAVs that ensures that users maintain proper coexistence with existing manned aircraft in the airspace (European Drone Regulations | Parrot Drones, 2024).

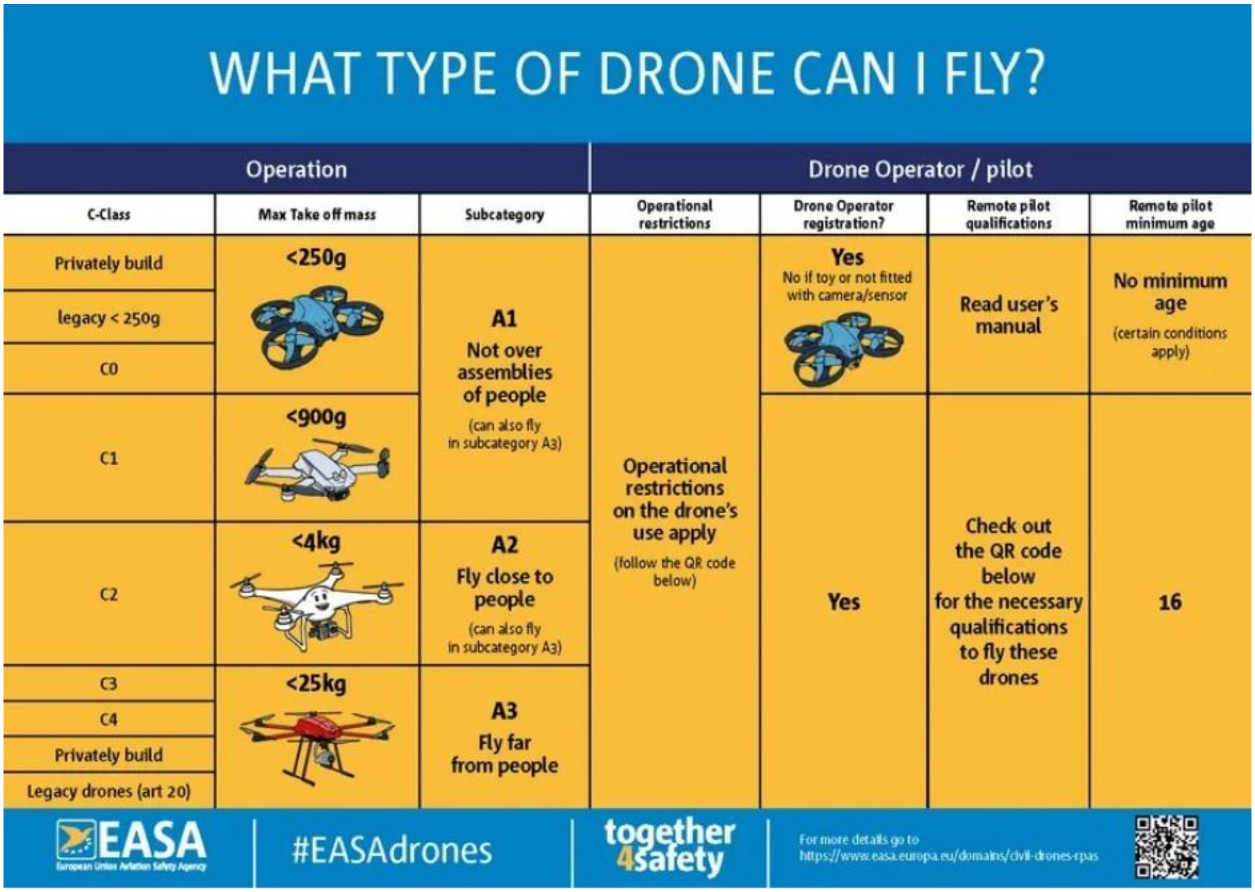

Figure 3: Requirements and limitations of different classes of drones (Drones (UAS) 2024)

2.2.5: Non-EU Manufacturers Regulation (EU) 2019/945 Requirement

Unlike the United States, the road map to certification for manufacturers from non-EU countries like Taiwan is more streamlined. There is no major barrier to entry into the EU UAV market that EU manufacturers do not already face. The same conformity assessment identified in Regulation (EU) 2019/945, which EU manufacturers are required to follow, also applies to non-EU manufacturers. However, because non-EU manufacturers lack a natural NAA to which they submit documentation to obtain a declaration of compliance, non-EU UAV manufacturers need to find an authorized representative. Unfortunately, because the 2019/945 regulation defines an authorized representative as "any natural or legal person established within the Union who has received a written mandate from a manufacturer to act on his behalf in relation to specified tasks," non-EU manufacturers need to identify and submit a written mandate to a legal representative who resides in any of the member states. As an established organization that serves the EU market with testing and certification services for electronic products, not just UAVs, the ETC can address this gap by exploring strategic partnerships with existing contacts or establishing new contacts to provide conformity assessment services for Taiwanese manufacturers. Unfortunately, our research did not yield any results for an easily attainable mandate template or information about existing authorized representatives that could be provided to the ETC in our recommendation. Therefore, the ETC might need to contact EASA for possible opportunities or resources to explore this path.

Beginning in February 2026, the European Union launched the Action Plan on Drone and Counter Drone Security to create a unified approach to guard against threats posed by malicious drones (European Commission, 2026). The plan focuses on the civilian internal security dimension while supporting civil-military synergies. For drone operators in the EU, this initiative has shifted perceptions of UAVs from primarily innovative tools to potential security liabilities.

The Action Plan aims to protect critical EU infrastructure, strengthen border security, and counter hybrid threats, positioning drones as a central element of the EU's defensive strategies (Joslyn, 2026). Technically, the plan emphasizes strengthening preparedness and detection capabilities, and coordinating responses from the military, thereby enhancing the EU's defensive readiness (Action Plan on Drone and Counter Drone Security, 2026). Additionally, the plan strengthens telecommunications and sensory components of UAVs in EU airspace, granting authorities greater oversight of drone connectivity and communications (TECH DRONE MEDIA, 2026). With heightened concerns around safety and security, combined with new strategic measures, the EU appears strongly focused on advancing UAV development in areas such as cybersecurity and connectivity, ensuring that drones in the European market meet both innovation and resilience requirements.

Given the influence of these two major economies on the global regulatory landscape, the development of the UAV industry in the U.S. and the EU is of great interest to the Taiwan Testing and Certification Center (ETC). As Taiwan seeks to expand its UAV sector and strengthen its global presence, ETC's engagement with the regulatory framework relevant to its services is particularly important to its efforts to establish a secure, trusted supply chain.

2.2.6: Taiwanese Certification

Taiwan has emerged as a strategically important country in the global UAV supply chain, especially as demand for trusted supply chains continues to grow. The Taiwanese government has invested heavily in domestic UAV development to strengthen its position in the global market.

Taiwan's UAV testing and certification programs are regulated by several organizations, including the Civil Aviation Administration (CAA), National Communications Commission (NCC), and Bureau of Standards, Metrology, and Inspection (BSMI). The CAA is a government agency, similar to the United States' FAA, responsible for regulating all operations in Taiwanese airspace (CAPA, n.d.). The NCC has a responsibility similar to that of the U.S. FCC for certifying devices that emit radio signals. This includes ensuring that UAV communication systems comply with Taiwan's radio frequency standards. Lastly, the BSMI is a government agency that develops national standards, conducts product inspections, and ensures metrology accuracy. They enforce safety regulations, electromagnetic capability (EMC), and quality testing for electronics (JJRLAB, 2024)

The Industrial Technology Research Institute (ITRI) (like ETC) is a non-profit applied research institute established with government support. ITRI has partnered with government and international organizations, such as the Association for Uncrewed Vehicle Systems International (AUVSI), to improve the UAV certification process and help Taiwan's drone manufacturers gain a larger share of the non-China supply chain (ITRI, 2019; AUVSI, 2026).

In order for Taiwan to achieve its long-term goals of creating a resilient, competitive, and internationally trusted UAV industry, this research project's core challenge was to address the fragmented certification pathway that impedes the production of drones for both domestic and international markets. Our core objective was to establish an efficient certification and reliability testing regime for UAVs—radio frequency communication and other related UAV technologies—that aligned with U.S. and EU regulations.

To achieve this strategic objective, four separate results were achieved:

1. Standardization of certification pathways. Processes become transparent and simplified, resulting in reduced uncertainty, improved planning capability, and lower compliance costs for manufacturers.
2. Testing delays are being reduced. By identifying problems within the system and through the recommendation of improvements that align with industry and government demand, future production speed could be increased (Harman, 2026).
3. Recommendations of harmonization strategies and potential pathways towards mutual recognition. Taiwan has already made advances toward international regulatory alignment. As of January 28, 2026, the ITRI is a member of the Green UAS certification program as a recognized cybersecurity assessor (AUVSI, 2026). Nevertheless, there is still room for improvement. Taiwanese UAV manufacturers continue to face export barriers because of the evolving FCC, FAA, and EASA standards. By conducting a comparative regulatory analysis and identifying duplicative or misaligned requirements, the project was able to recommend a way for testing agencies and manufacturers to work together to increase certification output.
4. Strengthening of institutional connections. Effective certification depends on technical standards as well as coordination between the ETC, regulatory agencies, manufacturers, and international partners. By improving communication and collaboration, as in the case of ITRI and AUVSI, fragmentation is reduced, and supply chain trust improves. Stronger connections create shared incentives for sustained system improvement.

Although Taiwan has made significant progress in developing UAV technologies, the gap between regulatory and international standards persists. Differences in certification pathways, documentation requirements, and even shipping standards create inefficiencies and limit global competitiveness. This highlights the need for an internationally standardized certification framework, which this research project focused on.

2.3: Additional Research

To touch on more relatable topics in drone development, this research examined performance testing, BVLOS operations, and Remote ID technology in both the U.S. and EU as they expand their UAS frameworks. We conducted this investigation to gain a more complete understanding of UAV technological development that will inform our recommendations to the ETC.

2.4: Performance Testing

To cover the entire regulatory and certification process of UAV development, product testing is conducted by certification bodies like ETC, which focuses mostly on certification testing to ensure components are in compliance with essential safety standards. This type of testing commonly involves meeting regulatory requirements and conducting safety testing to ensure products are safe for both the environment and the people living in it. However, another type of testing evaluates how systems behave under various operational workloads to ensure they perform as designed. Although most often referenced in the context of software development, this type of test evaluates metrics such as speed, stability, scalability, and reliability of the product being tested (Narwhal, 2018; Cohen, 2022). Similar metrics are also evaluated for UAVs by manufacturers and research institutions.

2.4.1: United States Performance Testing

In 2018, the U.S. National Institute of Standards and Technology (NIST), an organization under the U.S. Department of Commerce, developed, and published a guide of over 50 testing methods that can be used to assess the performance of UAVs weighing less than 25kg (55 lbs) at takeoff (NIST, 2018). Together with organizations such as the Airborne Public Safety Association (APSA), the National Fire Protection Association (NFPA), and the American Society for Testing and Materials (ASTM), these methods are being standardized for industry, agencies, and individual pilots to adopt easily.

Performance tests are divided into five levels. Levels 1—3 describes Open Test Lane flights (flying with no obstacles or restrictions in the flight path), in conjunction with a Basic Safety Check flight, Maneuvering Only, and Payload Functionality, respectively. Level 4 describes Obstructed Testing Lane (Flying with obstacles in the flight path) and Level 5, Confined Test Lane (flying in a confined space, such as in a building (Levels of Proficiency Introduction, NIST, 2024). Although an official certification process has been developed for passing these established levels, anyone can take the test inexpensively using free presentations provided online by NIST (Levels of Proficiency Introduction, NIST, 2024).

If official certification is desired by taking the tests, NIST has approved APSA to administer an official certification for Basic Proficiency Evaluation for Remote Pilot (BPERP), which is only based on Level 1, Open Test Lanes, evaluating performance through five maneuvering exercises within a designated ten-minute time limit (Level 1 Basic Proficiency Trial | NIST, 2024). Evaluation for this certification process must be administered by an APSA-approved proctor (NIST / APSA Certification; UAS Basic Proficiency and Advanced Proficiency - SUAS PRO, 2022), or through APSA. Regardless, to be eligible, the pilot must have successfully completed the BPERP flight evaluation administered by an APSA-approved proctor with a minimum score of 80% (or 32 in 10 minutes), apply for the certificate, and pay the fee upon passing to APSA (APSA Member Portal, 2025).

Fortunately, APSA didn't just stop providing BPERP testing; it also provides services to qualify proctors and instructors through its train-the-trainer program. This program is a 3-day course that provides BPERP and Advanced certifications, qualification as a designated proctor, and, if completed three times, eligibility to become an instructor (Murphy & Murphy, 2022).

After research was conducted into APSA's train-the-trainer program, no valid results were obtained from APSA regarding how participants get involved or the required materials, except for two official course postings in 2022 and 2024, shown in the respective figures (Figure 4 and Figure 5). The 2022 Advanced course in Florida was listed at $575, while the 2024 Advanced/Confined course in Maryland was listed at $675. Together, these two tests, although limited, gave an idea of the Train-the-Trainer cost of around $625. Regardless of this information, contacting APSA to obtain accurate details on how to follow the program roadmap to become a proctor/instructor for NIST performance testing is needed due to the limited information we have gathered online.

NIST Performance testing is becoming standardized across industries and around the world with activities from early adopters like the National Fire Protection Association (NFPA) and ASTM International, for assessing drone performance but also in training pilots.

NFPA has adopted this methodology for Public Safety Small Unmanned Aircraft Systems (sUAS) programs, especially with the NIST testing referenced in the NFPA Job Performance Requirements section in the National Fire Protection Association Standard for Small Unmanned Aircraft Systems Used for Public Safety Operations (NFPA 2400) (NIST, 2020). As a supplement to this, there are more applications of these methods in emergency response activities and public safety agencies because of the great potential for their use in ground and aerial to aquatic operations (NIST, 2025).

Meanwhile, ATSM International, a globally recognized leader in the development and delivery of voluntary consensus standards, similar to organizations like ANSI, ISO and IEC, has created a Response Robots (E54.09) Report that standardized more than twenty test methods created by NIST (Henderson, 2023; Pilot Institute, 2021) and references NIST tests in the ASTM Standard Guide for Training for Remote Pilot in Command of Unmanned Aircraft Systems Endorsements.

Together, NFPA and ASTM have begun paving the way for NIST performance testing guidelines to serve not only the U.S. market but also the global market. Through this initiative, the ETC can also explore a U.S.-trusted method to inform their operations and increase collaboration with Taiwanese manufacturers for U.S. markets.

NIST Advanced sUAS Standard Test Methods; Proctor Training Course

DATES:	27-29 October 2022 (0800-1800)
LOCATION:	Northwest Florida State College Criminal Justice Training Center 100 E. College Boulevard, Building 400, Room 132 Niceville, Florida 32578
COST:	$575.00
RESERVATIONS:	Airborne Public Safety Association Telephone: (301) 631-2406 Web Site: https://publicsafetyaviation.org/2022-on-the-road-nist-advanced-suas-standard-test-methods-proctor-training-course-niceville-fl

COURSE DESCRIPTION: 24 hours of classroom and hands-on flight instruction covering advanced aspects of the National Institute of Standards and Technology sUAS Standard Test Methods. The course will address constructing and managing the NIST obstructed test lanes, night operations, beyond visual line of sight (BVLOS) operations, and embedding apparatus within scenarios. These test methods can be used to evaluate sUAS capabilities and sensor systems, or remote pilot proficiency for credentialing. The tests are easy to conduct alone or in groups, and inexpensive enough to set up multiple concurrent lanes. They are quick to perform, typically less than 30 minutes to conduct all the tests in a given lane, so they can support flying practice for remote pilots at the beginning of every training session. The NIST sUAS Standard Test Methods are an excellent way to add a sUAS pilot flight skills credentialing component to your sUAS program. NIST has created a comprehensive user guide, scoring forms, and apparatus targets that can be printed and placed in the test apparatus buckets. Attendees will learn how to fabricate apparatus, conduct trials, and embed them into their own training and credentialing programs. The NIST sUAS Test Methods have been adopted, or are under consideration for adoption, by the Airborne Public Safety Accreditation Commission, National Fire Protection Association, Civil Air Patrol, and ASTM International.

Attendees should be experienced sUAS pilots who want to hone their skills, evaluate sensor systems and/or have a desire to train and evaluate other sUAS pilots. Ideally, they will have previously completed the APSA NIST Basic sUAS Standard Test Methods Proctor Course. Attendees must bring their own quadcopter style sUAS, capable of at least 15 minutes of flight time, equipped with a camera and anti-collision lighting. Additional sUAS batteries and a battery charging station are also required. Self-contained illumination mounted on the drone (LumeCube or similar), and a laptop computer are highly recommended.

NOTE: A minimum of 8 paid registrants by October 6, 2022 is required to conduct this course.

Updated 02/07/2022

Figure 4: APSA NIST Proctor Training Course at Criminal Justice Training Center (NIST SUAS Standard Test Methods 2022)

sUAS Proctor Training Course

ADVANCED/CONFINED

Based on NIST Aerial Drone Tests and Scorable Scenarios

DATES: 20-22 May 2024 (0800-1700)

LOCATION: **Randallstown Community Center**
3505 Resource Drive, Randallstown, MD 21133

COST: $675.00

REGISTRATION: Airborne Public Safety Association
Telephone: (301) 631-2406
Web Site: https://publicsafetyaviation.org/events/uas-training/2024-on-the-road-nist-suas-standard-test-methods-proctor-training-course-advanced-confined-randallstown-md

COURSE DESCRIPTION: 24 hours of classroom and hands-on flight instruction covering the National Institute of Standards and Technology (NIST) sUAS Test Methods - Test Lanes Levels 1-5: Open Test Lane Level 1 - Basic Proficiency Evaluation for Remote Pilots (BPERP-Part 107 qualification); Open Test Lane Level 2 - Maneuvering; Open Test Lane Level 3 - Payload Functionality; Obstructed Test Lane Level 4; and Confined Test Lane Level 5.

The NIST sUAS Test Methods are an excellent way to add a sUAS pilot flight skills credentialing component to your sUAS program. NIST has created a comprehensive user guide, scoring forms, and apparatus targets that can be printed and placed in the test apparatus buckets. Attendees will learn how to conduct trials, and embed them into their own training and credentialing programs.

Attendees must have their Part 107 license and must be experienced sUAS pilots who want to hone their skills, evaluate sensor systems and/or have a desire to train and evaluate other sUAS pilots. Attendees must bring their own quadcopter style sUAS, capable of at least 15 minutes of flight time, equipped with a camera. Additional sUAS batteries and a battery charging station are also required.

Successful completion of this course will provide you with:

- APSA Open, Obstructed and Confined Test Lanes (Levels 1-5) Remote Pilot Certificate
- APSA Open, Obstructed and Confined Test Lanes (Levels 1-5) Proctor Certificate. This will allow you to serve as a proctor for all test lanes.

NOTE: All attendees must be registered and paid in full to attend this course.

Updated 03.22.2024

Figure 5: APSA NIST Proctor Training Advanced/Confined Course with Texas Department of Public Safety (sUAS Proctor Training Course Advanced | Confined 2024)

2.4.2: EU Performance Testing

In Europe, a notable development redefining the performance testing landscape is the National Experimental Test Center for Unmanned Aircraft Systems at Magdeburg-Cochstedt Airport, developed by the Deutsches Zentrum für Luft- und Raumfahrt (DLR).

This testing center is housed in an active commercial airport in Saxony-Anhalt, Germany, which enables holistic technology development and serves as a pioneer for future UAV research and development led by DLR and its external industrial partners. In addition to the unique location in which this testing center is housed, at the end of 2025, a new GeoZone was established for the Cocstedt test area, which accelerates flight test activities at the center.

The Geozone opened up approximately 16 by 5 kilometers of new airspace, which gives test area engineers the ability to conduct test in Cochstedt without having to apply for otherwise necessary operational approval in the "specific" category, saving time and resources in preparing test flights (Drone Testing— from Everyday Applications to Defense, 2025).

At the time of this research, multiple projects and test activities have led to many developments in the German commercial defense industry. Four examples of development and testing at the DLR site include:

- The DLR partnership with EnBW (Energie Baden-Württemberg AG), one of the largest energy supply companies in Germany and Europe on the Offshore Drone Challenge.
- Test flight of a Volodrone cargo drone in traffic scenarios as part of the European-funded CORUS-XUAM project.
- Test flights of the DLR ALAADy-CC cargo drone demonstrator.
- Partnership with the start-up company Labfly on trials of a medical logistics drone.

All four of these tests involved multiple types and classes of drones for different applications.

DLR and EnBW's collaboration on the Offshore Drone Challenge, held on June 19 and 20, 2024, featured four companies (ADLC, Stromkind, Solectric, and Flowcopter) showcasing their skills in competitive flights on airport grounds. ADLC operated the "orca" drone from manufacturer Pheonix Wings, Flowcopter, and Stromkind, operating DJI drones and Solectric, presenting their own drone technologies. Each company was evaluated on its UAV's maneuvering capabilities. (EnBW and DLR Organize Drone Competition for Offshore Wind Farm Use, 2024). Each drone was tasked with the operation and maintenance logistics of offshore wind farms with specific assignments to pick up, transport, and set downloads using as much automation, wind farm communications, and efficient navigation as possible in a BVLOS situation.

The result of this challenge spurred discussion of future use of drones in the offshore industry, with discussion being not only technical, but also regulatory in nature (EnBW, n.d.). As a result, a key validation step was created for the broader Upcoming Drones Windfarm (UDW) project that ended in April 2025 with the development of a holistic system architecture for autonomous drone operations, which emphasizes onboard mission management, contingency planning, and robust communication networks (Schopferer et al., 2025). This highlights what testing the ETC should focus on regarding UAVs and the pathway into the EU market.

The second project at the National Experimental Center introduced in our research was the test flight of the VoloDrone cargo drone in traffic scenarios, conducted to demonstrate deconfliction and aircraft spacing procedures, ensuring that electric Vertical Takeoff and Landing (eVTOL) aircraft can seamlessly fly amongst established aircraft such as rescue helicopters. This test was part of the European-funded CORUS-XUAM, which focuses on integrating digital Air Traffic Management (ATM) and Unmanned Traffic Management (UTM) interfaces to enable new airspace pilots to safely operate in existing airspace. The test included an ADAC rescue helicopter as a manned aircraft with which the VoloDrone UAV interacted, using Volocopter's VoloIQ digital ecosystem, working in tandem with a U-Space service provider (USSP), to manage flight paths and speeds in congested airspace.

As part of the flight test, the VoloDrone's flight path between Messe Frankfurt and Frankfurt Airport was diverted due to an incoming signal from an EMS aircraft's priority landing at a hospital enroute to the VoloDrone UAV location (Rees, 2022). This contributed to the flight test's success. In addition to the results of these tests, several improvements to Volocopter's Volodrone were made to help the company work towards its goal of entry into commercial service. Overall, this highlights the development of UAV integration into airspace management controls and systems, which might affect regulatory standards and specifications that inform the ETC as progress is made to support Taiwan's entry into the EU UAV market.

The third testing case with DLR was a successful test flight of the ALAADy-CC (Automated Low Altitude Air Delivery – Cross Country) cargo drone demonstrator. This test involved a 450-kilogram gyrocopter undergoing extensive modifications, including the installation of a freight-carrying area and custom actuators. This test was intended to investigate the transport of larger cargo items by gyrocopters, a form of low-level air transport. As part of the findings, the freight UAV reached an altitude of up to "150 meters and traveled at speeds of approximately 100 kilometers per hour" with theoretical research completed on payloads of up to one tonne (Ball, 2019). Current developments in this area include the development of UAV automation for flight procedures, integration into standard airfreight chains, clarification of legal constraints, and risk assessment before entering the market (DLR Conducts Flight Tests for Gyrocopter Drones, 2019). An early concept for this type of technology could be explored by the ETC using current knowledge from established electronic devices for which they provide tests, and by exploring alignment with the EU's SORA assessment, adopted by the EASA for the complex category in which this type of UAV will operate.

Lastly, covering DLR trials with Labfly, a medical logistics drone startup, which conducted a test in 2022 to evaluate speed and contactless delivery performance. As the first drone operator in Germany to allow transport of dangerous goods above populated areas and beyond visual line of sight, the test had real-world implications that have allowed Labfly to complete over 1,500 certified flights with reduced delivery time from 90 to under 15 minutes and cutting overall costs by 60% (Labfly, 2025; Green Health Accelerator, 2026). This highlights the impact drone startups are having on the commercial market; therefore, ETC should continue working with startups.

2.5: Beyond Visual Line of Sight (BVLOS)

In the context of UAVs and this research, Beyond Visual Line of Sight or BVLOS is used as a verb. BVLOS is defined as operating in an area that is beyond the remote pilot's direct visual line of sight or a situation when the pilot is unable to maintain continuous unaided visual contact with the UAV (SKYbrary Aviation Safety, n.d.). This type of operation, although it introduces higher risks and management concerns when operating UAV devices, enables full scalability and large-scale operations for which drones are now in demand across many industries. BVLOS operations are enabled by more sophisticated technologies such as command-and-control links, network connections, sensors, navigation systems, and onboard intelligence. As a result, different frameworks are working to properly outline how these operations would work.

2.5.1: United States BVLOS

As mentioned in section 2.2.2, the FAA controls regulations for drones in the United States, defining how UAVs are operated. It also has the authority to define U.S. UAV Beyond Visual Line of Sight (BVLOS) operations. Under 14 CFR Part 107, which defines regulations for commercially or recreationally used drones, BVLOS operations currently require a waiver or individual exemption directly from the FAA. At the time of this research, failing to obtain a waiver or exception before a BVLOS operation was illegal (Stoner, 2026). This is an inefficient and time-consuming process for pilots. Fortunately, as part of the development of 14 CFR Part 108, a new regulatory framework is designed to standardize a performance-based system for routine BVLOS operations, which will replace the current case-by-case waiver process.

As of August 2025, a Notice of Proposed Rulemaking (NPRM), a formal announcement by a U.S. federal agency like the FAA that proposes new or revised regulations and invites public comment before finalizing the rule (Eye, 2026 and French, 2025). The NPRM sought to "propose performance-based regulations for the design and operation of UAS beyond visual line of sight (BVLOS) and for third-party services that support these operations" (Federal Register, 2025). Although the original public comment period was set for 60 days from August 7 to October 6, 2025, during which the FAA received over 3000 comments from industry operators, stakeholders, interested/concerned citizens, and news organizations, the NPRM was reopened for more comments during a public 14-day comment period from January 28 to February 11, 2026. The purpose of reopening the issue was to gain a greater understanding of three specific topics: electronic conspicuity (EC), right-of-way paradigms between drones and crewed aircraft in low altitude airspace, and detect and avoid technology exception in high-risk environments, such as Class B/C airspace and over densely populated areas (Magdolna, 2026). Based on the typical rulemaking timeline, multiple sources predict that the final rule will be published in the U.S. Federal Register in spring 2026, with implementation 6—12 months after publication. Although not finalized at the time of this research, a preliminary NPRM can be viewed online through the Federal Register.

2.5.2: EU BVLOS

As demonstrated in section 2.2.4, the EU uses a safety framework which defines categories and classes of UAVs based on risk level and operational intent. According to the regulation EU 2019/947, Beyond Visual Line of Sight (BVLOS) operations cannot be performed in the Open Category except in special cases known as extended Visual Line of Sight (EVLOS) – a situation where the operator/pilot's eyes are on the drone indirectly using visual observers or aids. However, drones in this specific category can perform BVLOS operations. This is due to the higher risk such an operation introduces (Mugo & Mugo, 2025). In addition to authorization and complaints, a pilot needs to operate a BVLOS operation in the Specific category mentioned in section 2.2.4. The UAV device requires a Remote ID and has a Detect-and-Avoid (DAA) system.

2.6: Remote ID

Remote Identification (Remote ID) has become a pivotal part of current UAV regulation. The system allows a UAV to broadcast identifying information while in flight; essentially a digital license plate for drones. Instead of requiring authorities to physically recover or inspect a drone after an incident, the system allows certain information to be received remotely during use. This can include information like the aircraft's identity, location, altitude, speed, and information connected to the operator. As UAVs become more common, Remote ID has grown to be an important part of drone regulation because it supports accountability and safety in the long-term integration of drones in shared airspace (Federal Aviation Administration, 2025; European Union Aviation Safety Agency, 2023).

Although the U.S. and EU both require Remote ID, they apply it differently through their own regulatory structures. In the U.S., Remote ID is mainly handled by the manufacturer and then overseen by the FAA. Manufacturers are required to show that their aircraft or broadcasting module satisfies federal performance requirements before it can be accepted as compliant for operation in the U.S. airspace (Federal Aviation Administration, 2024). In the EU, Remote ID is incorporated into their risk-based system that connects drone class labels and operating categories to technical requirements (European Union Aviation Safety Agency, 2023). Overall, Remote ID has the same purpose in both regions but is implemented through different legal and technical pathways.

2.6.1: U.S. Remote ID

In the U.S., Remote ID is regulated by the FAA as part of the federal rules governing UAVs. The FAA requires that certain drones be built with either built-in Remote ID capability or a separate broadcast module, both of which must meet a set of performance requirements before it can be certified for use in the U.S. (FAA, 2024). FAA requirements describe what the system must do, such as broadcasting identification and location information, rather than requiring a singular hardware design. This allows manufacturers to use different hardware, as long as they can show the final product broadcasts the required information (FAA, 2025).

To show that a product meets the FAA's requirements, the manufacturer must use an accepted (FAA) procedure to demonstrate compliance. This must include hardware testing and supporting documentation, to show that the product satisfies Remote ID requirements. The manufacturer also submits a formal record to the FAA called a Declaration of Compliance (DOC). This document is an official statement from the manufacturer stating that the drone or broadcast module meets Remote ID performance requirements. If the FAA accepts the submission, the product is recognized as in compliance for use in the U.S. (FAA, 2020).

This system shows that the FAA does not treat Remote ID as a label or marketing claim but instead uses a structured process requiring documented technical support and formal submission. The manufacturer remains responsible for ensuring that products listed under an accepted declaration continue to conform to the approved design and performance requirements. FAA guidance also says that the agency may inspect products, review records, and revoke acceptance if a listed product no longer meets standards.

2.6.2: EU Remote ID

In the EU, Remote ID is implemented through a broader regulatory system that connects technical requirements to both the class of the drone and the risk level of the operation being performed. Starting on January 1, 2024, all drones operating in the specific category, as well as all drones with class marks operating in the open category, are required to use an active and up-to-date remote ID system (EASA, 2023).

In the open category, Remote ID is associated with class marked drones C1, C2, C3, C5, and C6. These classes are connected to different operating situations based on risk and proximity to urban environments. Class C4, however, is treated differently because it is intended to be a class for more traditional model aircraft which do not normally carry the same built-in Remote ID requirements as the other classes. The EU rules also provide requirements for a separate Remote ID module in situations where it may not be built into the drone.

This structure reflects the EU's risk-based approach for drone regulation where requirements depend on both the design and purpose of the drone. Rather than applying the same identification rule to every drone, the EU connects Remote ID to product class, operational risk, and operator responsibilities (EASA, 2023). The EASA also states that the operator registration number must be uploaded into the drone's Remote ID system, making sure that when in flight, the drone is able to be traced back to a registered user in the system (European Union Aviation Safety Agency, 2020). Remote ID in the EU is part of a larger regulatory framework that connects the drone, operator, and intended use, helping create a clearer and more consistent system for oversight and market access.

Chapter 3: Methodology

3.1: Research Components

With each country having its own specific regulations for drone manufacturing and use, this research project required us to gather information on testing, certification, regulatory, and export guidelines. Through this work, we were able to provide recommendations to both ETC and two manufacturing companies, focused on aligning these companies' technical standards with global standards identified in our research. Alignment with international standards is the basis for getting Taiwan on track to become a part of the global UAV industry.

The initial data requirements are all information pertaining to regulations, testing, certification, and export practices in Taiwan, the United States, and the European Union. With a clear understanding of this information, we were able to create a website for the ETC that is clear, easy to understand, and easy to navigate.

In addition to these testing and regulatory information, we also determined the potential market-entry strategy for the ETC to take its first step in expanding its UAV testing and collaboration capabilities. There is a wide range of growth and collaboration opportunities for the ETC, from taking the step to becoming a Green UAS certifier to initiating domestic and international collaborations between Taiwanese and other global manufacturers.

3.2: Data Collection Methods

Data collection for our research project involved two primary methods: literature reviews and background research, and interviews/presentations from ETC staff members. To gain a stronger understanding of Taiwanese drone regulations and the ETC's current testing infrastructure, we conducted interviews and discussions with subject matter experts (SMEs). In addition to these interviews, ETC staff members delivered presentations on their areas of expertise, including electromagnetic compatibility (EMC), radio frequency (RF), and related testing domains. These presentations provided the research team with deeper insight into the ETC's specific testing requirements, certification procedures, and the technical considerations involved in evaluating UAV systems.

To understand the requirements of the EU and the United States, we conducted extensive literature reviews of the regulations and requirements surrounding their respective drone industries. Specifically, our research focused on the regulatory structure, product certification requirements, import requirements, and compliance systems. This research is detailed more in our background research section.

This mixed-methods approach was appropriate for this project because no single source would fully explain both the regulatory and certification practices of all the certification bodies. An extensive literature review provided the external regulatory context for the European Union and the United States, while the interviews and staff presentations provided internal insight into the ETC's capabilities, limitations, and current operations. Together, these methods created a stronger basis for analyzing certification gaps and developing recommendations that are both internationally informed and practically applicable to Taiwan's UAV industry.

3.3: Questionnaires

Perspectives and personal experiences from experts directly involved in the various facets of the testing and certification process at ETC greatly benefitted the development of a website and recommendations tailored for ETC use. Our overall objective on this questionnaire was to gain greater insight into the current state of each subject's testing practices and the direct work being done. All respondents were asked to sign a consent form, which is in Appendix A, full transcripts and notes from the questionnaires/interviews are in Appendix B.

3.3.1: Questionnaire with Subject Matter Experts

The questionnaire was sent to a total of 6 subject matter experts in different areas of UAV testing and certification services. These topics included cybersecurity, reliability, EMC, geofencing, certification, and communication. Following the questionnaire design, respondents were not restricted in how they answered these questions, responding with as little or as much information as possible. This unstructured format introduced uncertainty in the information provided by experts and in what they focused on in their responses.

For the topic of drone safety, we received a response from subject matter experts (SME) Andre Pan, the ETC's lab manager, which focused on assessing current ETC capabilities, guidelines, and challenges related to drone safety.

For the topic of drone reliability, we received responses from SME Mike Tsai, ETC's leader in the Environmental and Reliability Testing Division. Here, the respondent focused on issues the ETC faces in reliability testing for drones and potential solutions using standards the ETC currently uses or references.

For the topic of UAV EMC and geofencing, we received responses to the questionnaire and interviewed subject-matter expert Eric Chen, the ETC's manager of the EMC Testing Department No. 1. In both the questionnaire and the interview with Mr. Chen, the respondent focused on specific Taiwanese standards applied in the current ETC's EMC and geofencing drone testing, as well as the challenges in scaling testing to international markets.

Under the topic of UAV certification, the questionnaire received responses from SME Ariel Hsieh, the ETC's sales representative responsible for working with ETC's customers. The respondent focused on international standards that the ETC references for drone detection, as well as gaps the ETC faces in drone certification, as they look for ways to provide not just testing services but also certification services to their customers, i.e., drone manufacturers.

For our questionnaire on the cybersecurity of UAVs, we received responses from Dr. Hsi-Hsun Yeh, the ETC's senior manager of the Information and Communications Technology Service Department. The response to this questionnaire focused on Taiwan's 2024 UAV cybersecurity regulation, titled the 2024 Remote Drone Capital Security Specification Testing, ETC's Taiwan Accreditation Foundation (TAF) accreditation, and the current focus of cybersecurity tests.

Lastly, regarding UAV communications, we received responses from Dr. Roy Lin, a manager in the ETC's Information and Communications Technology Service Department. Dr.

Lin's response focused on a specific communication standard that could enable ETC UAV testing in this area, as well as on some ETC research findings in this area.

3.4: Recommendation Analysis

By combining information gathered from questionnaire responses, notes from lab tours, research findings, and other materials from UAV manufacturer site visits, our team developed a data-driven recommendation for this study. We also conducted thematic and comparative analyses. The primary Python libraries the team used include sklearn.feature_extraction.text, pandas, regex, os, and rapidfuzz. The two analysis code files will be available in the GitHub directory that was provided to the ETC, along with other code files for future analysis generation.

3.4.1: Thematic Analysis

The thematic analysis began by importing two datasets:

1. ETC-sourced dataset, containing UAV testing information categorized into areas such as reliability/airworthiness, cybersecurity, and radio frequency. This data was stored in an Excel file with columns (doc_id, title, source, category, doc_type, summary).
2. Online dataset, compiled from external sources, containing regulatory and standards-related information. This dataset included columns headings such as Region, Primary Category, Regulation/Certification Topic, Source Title, Brief Summary, Article URL, and File Name/Origin.

To ensure consistency, all column headings in the online dataset were standardized by trimming whitespace and converting text to lowercase; conditional logic was applied to assign categories based on the presence of predefined keywords.

The online dataset was reformatted to align with the ETC dataset structure, which was achieved by mapping existing columns to a unified schema (e.g., Source Title changed to title, Region became source, Primary Category became category).

Both datasets were further standardized by normalizing category labels and then combining them into a single dataset via concatenation. Rows with missing summaries were removed to ensure data quality, and text was converted into a numerical representation using TF-IDF vectorization.

With this numerical representation, calculations became possible, and term importance was determined and ranked based on frequency and relevance. For each category, all associated summaries were aggregated, and the most representative keywords were identified. These keywords were interpreted as the dominant themes within each category, as seen in Figure 5. This process was performed on the ETC-sourced and Online datasets for comparison with a bar chart developed to visualize results using matplotlib.pyplot tool.

3.4.2: Comparative Analysis

The comparative analysis focused on identifying alignment between standards referenced in the ETC dataset and those found in EU and U.S. regulatory datasets.

This process began by importing multiple datasets:

1. ETC-sourced dataset, containing UAV testing information categorized into topics such as reliability/airworthiness, cybersecurity, and radio frequency. This data was stored in an Excel file with columns doc_id, title, source, category, doc_type, summary.
2. Online dataset, compiled from external sources, containing regulatory and standards-related information. This dataset included columns such as Region, Primary Category, Regulation/Certification Topic, Source Title, Brief Summary, Article URL, and File Name/Origin.
3. 8 Topic-specific online compiled data. 4 topic-defined datasets: Cybersecurity, EMC/RF, Reliability, and Safety, contained information on European technical standards found online, while the other 4 datasets focused on the United States. These datasets were titled using the convention region_topic_docs.

Using the Python regular expressions tool, patterns were identified corresponding to known standard formats, e.g., IEC, ISO, EN, etc., and individual standards were extracted. These extracted standards were normalized by converting them to lowercase, removing excess whitespace, and applying processes such as replacing inconsistent dashes. A similar matching method was implemented using fuzzy string matching (rapidfuzz) to determine whether two standards should be considered equivalent, with a 90% similarity threshold to ensure a high-level of similarity between ETC and online-sourced standards.

To organize the analysis, external files were structured by region and topic using a naming convention, e.g., eu_cybersecurity_docs.xlsx. File names were parsed to automatically identify the region and topic. Using this systematic parsing method, coverage was calculated at three levels, as a result with the first being at the topic level, e.g., U.S. Cybersecurity, followed by the region level, e.g., EU total, U.S. total, and then the global level, e.g., combined EU and U.S.

Finally, all results were exported into CSV files (Tables 1 - 4), including coverage summaries, matched standards, and unmatched standards (both directions). After reviewing the results of these analyses, we developed our recommendations to the ETC.

region	topic	total	covered	not_covered	coverage_pct
EU	communication	1	0	1	0
EU	cybersecurity	3	0	3	0
EU	ems	160	4	156	2.5
EU	resilience	3	0	3	0
EU	safety	3	0	3	0
U.S.	communication	4	0	4	0
U.S.	ems	9	2	7	22.22
U.S.	resilience	0	0	0	0
U.S.	safety	3	0	3	0
EU	TOTAL	168	4	164	2.38
U.S.	TOTAL	15	2	13	13.33
ALL	TOTAL	183	6	177	3.28

Table 1 – Standard coverage summary table from comparative analysis

3.5: Export and Regulation Control Interactive Interface

During this project, we have also developed an Export and Regulatory Control Interactive Interface to organize and present the regulatory and certification information collected from our literature review. The purpose of this is to make complex UAV-related regulations, testing requirements, certification procedures, and export control information more accessible and easier to navigate for the ETC and other relevant stakeholders. Rather than requiring users to search through lengthy documents or manually compare multiple regulatory sources, the interface provides a structured, interactive format that allows users to navigate information by area, regulatory body, and topic.

The interface is designed as a website with a hierarchical navigation structure, where users select a jurisdiction area, such as Taiwan, the United States, or the European Union. From there, they will be able to navigate to topics, such as Radio Frequency, Electromagnetic Susceptibility, or Airworthiness and Reliability. After choosing a topic, Users will be able to select categories such as regulations, testing requirements, certification pathways, and export-related controls. This structure mirrors the comparative framework used in this project, while allowing users to quickly locate the information most relevant to their search criteria and making it easier to identify different regulatory systems. Users can also quickly search for a specific topic, such as a specific drone certification, utilizing the search bar.

The content included in the interface was taken directly from our literature review, background research, and information gathered from the ETC staff for interviews and presentations. As the information was collected, it was distributed by region, regulatory authority, and requirement type before being transferred into a suitable format for the website. This process ensures that the website is not a separate product from the research, but an organized presentation of the findings in a more practical and user-friendly interface.

Chapter 4: Results and Analysis

4.1: Data Analysis Results

After conducting a comparative analysis of the questionnaires (see APPENDIX B) and our findings on U.S. and EU regulations pertaining to UAV technology, we identified gaps and relationships among ETC, U.S., and EU capabilities. Our analysis reveals very little overlap between ETC-referenced standards and those found in our research on EU and U.S. regulatory frameworks. As shown in Figure 6, only 3.3% of the EU and U.S. standards/specifications were identified among the testing capabilities that ETC shared with our team. One contributing factor is that the specific standards IEC 61850, CNS 16014, and CNS 15511-2 identified by ETC's drone communication SME in our questionnaire were not present in the online U.S.–EU standards found. This also highlights gaps in coverage for standards such as UL97, which the SME referenced in relation to drone certification. Although some of this low alignment is driven by differences in regional standardization systems, multiple in-development standards in both the U.S. and EU, and the amount of information gathered and extracted from accessible materials, there was a clear distinction between U.S., EU, and ETC standards. The results show that the ETC aligns with about 13.33% of U.S. standards, compared to 2.38% in the EU, suggesting that current capabilities support the regulatory frameworks in the U.S. market.

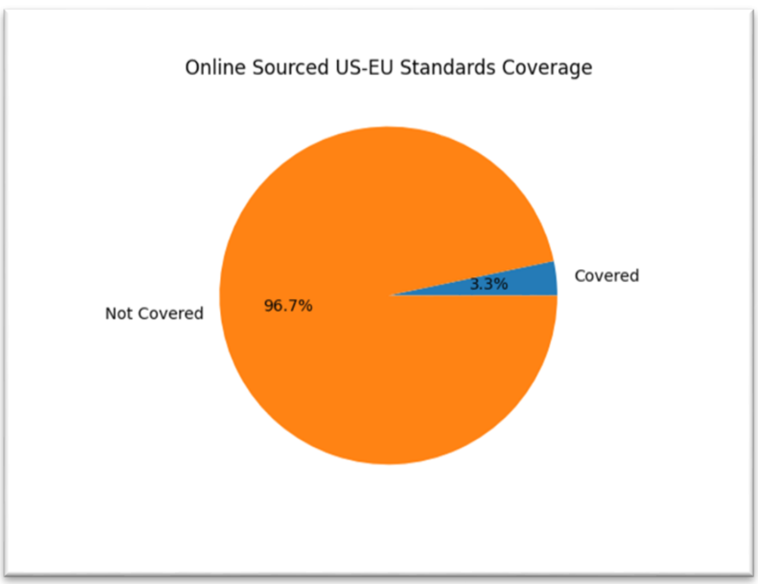

Figure 6: ETC and online sourced standard comparative analysis results

Lastly, the thematic analysis revealed a consistent focus between the ETC regulatory dataset and international online sources. As shown in Figure 7, among the four main topics (cybersecurity, EMC, reliability/airworthiness, and radio frequency) in which the ETC has an interest, three-quarters of those topics share similarities, with reliability/airworthiness being the only domain showing no similarity between the ETC-provided context and online-sourced context. A possible reason for this difference is the absence of such testing on drones at the ETC, coupled with the SME's emphasis on two standards IEC 60068 and MIL-STD-810 in their questionnaire responses for this topic. While these thematic domains are aligned across regions, the specific standards used to address them differ significantly, reinforcing the findings of low-direct standard overlap. With the two analyses indicating that ETC aligns more closely with the U.S. regulatory system in both thematic focus and standards usage, our recommendation would suggest that ETC serve as an assessor for Blue/Green UAV certification.

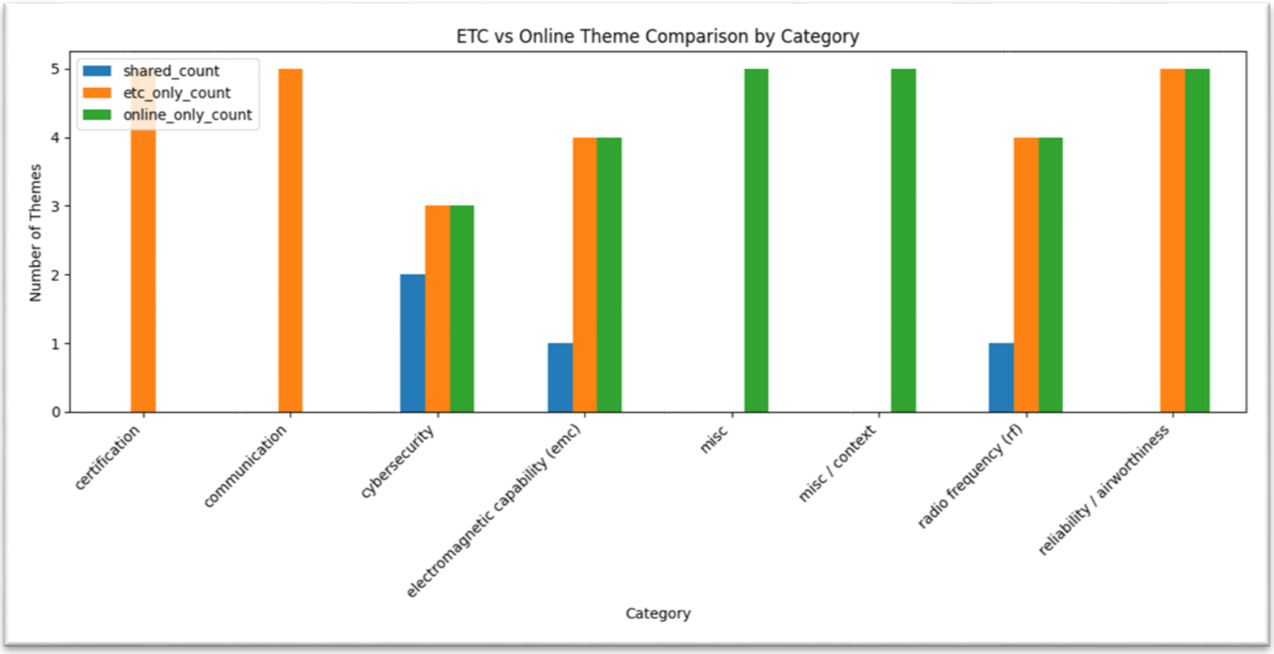

Figure 7: ETC and online sourced standard thematic analysis results

Additionally, with countries like Canada and India relying heavily on international standards shaped by EU and U.S. frameworks, ETC should prioritize alignment with widely adopted international standards by continuously monitoring the UAV regulatory landscape to track international standards and those referenced within emerging regulations. This should be followed by strengthening compatibility with U.S. regulatory capabilities and initiating cybersecurity testing based on EU-referenced standards, particularly in response to the EU's evolving digital security strategies for UAV systems. Fortunately, since the team's analysis was conducted using Python, future analysis can be generated using the same programs files.

4.2: Regulation Website Product Proposal

The primary goal of this product deliverable is to present a consolidated, easy-to-navigate interface that contains all the regulations, testing, certification, and export information identified through our combined literature review and on-site data collection. To do so, we used Python and Flask to develop a functional web-based prototype of the regulatory interface. There are three main aspects of the project that we utilized: User interface, navigation, and information addition.

4.2.1: Deliverable Scope

For the scope of this deliverable, the team focused on information delivery, user experience (UX), and long-term maintainability, rather than the website's visual aesthetics (UI). This product deliverable contains the project files, software components, structured JSON data used by the website, and supporting information spreadsheets used during research consolidation.

The program and software files focus on simplicity for future development and easy implementation on a deployed interface. By deploying the site on Render and maintaining the code on GitHub, the team created a product accessible via a public web link and transferred it to the ETC for continued ownership and improvement.

In the future, the ETC can expand this scope to support further developments in global regulation, including new laws, restrictions, and export regulations, as well as other regulatory categories. Further improvements in UI and UX can also be made for a full-fledged website.

4.3: Front End Design

The front end of the UAV Export and Regulation Interface was designed to provide a simple, hierarchical method for browsing regulatory information across multiple jurisdictions. Since the platform is intended to help users compare regulatory content by region, topic, and category, the interface emphasizes clarity, modularity, and ease of use over visual appearance. The frontend is implemented using Flask-rendered HTML templates and a shared CSS stylesheet, allowing the application to dynamically generate pages from structured regional JSON data, while maintaining a consistent visual layout across the whole site. The project structure separates application logic, content files, templates, and styling into distinct pages.

4.3.1: Page Structure

The page structure follows a layered hierarchy that follows the routing logic in app.py and the nested organization of the JSON files. Users begin at the home page, as shown in Figure 8, which displays all available jurisdictions. Selecting a region routes the user to a region page, as shown in Figure 9, where they can choose among major technical topics, such as RF, EMC, or reliability. From there, users proceed to a topic page, then a category page. Lastly, the subsection page contains specific content, while entries within that subsection contain specific information.

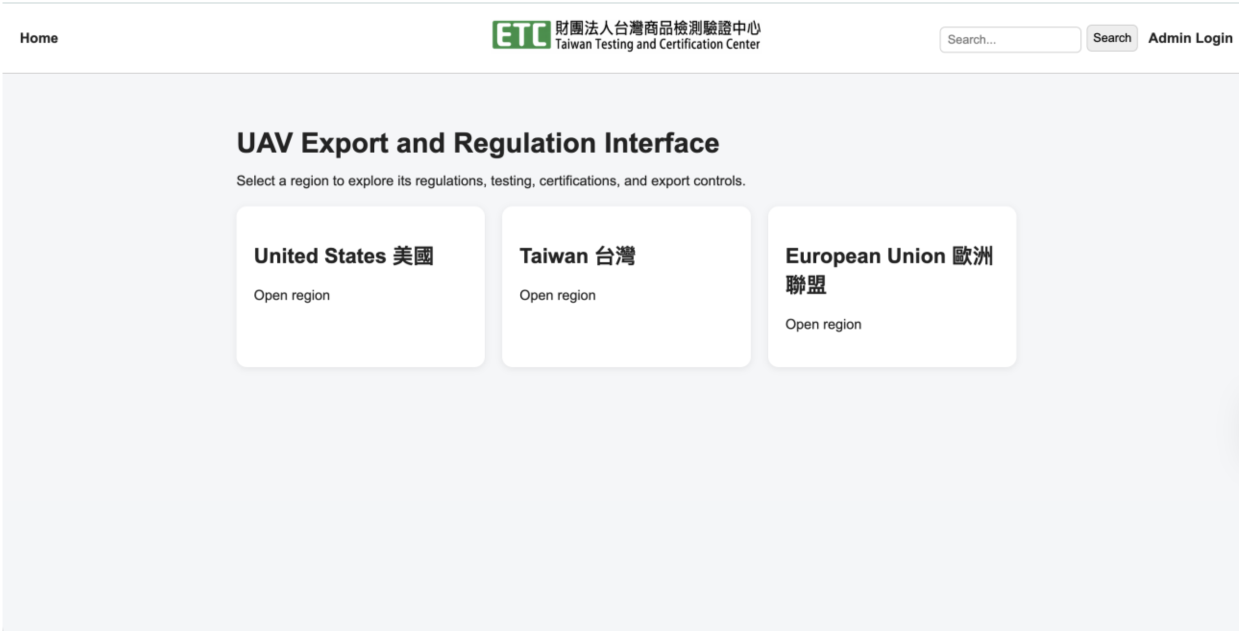

Figure 8: UAV Export and Regulation Interface (Website)

This structure was chosen because it matches the intended workflow of first selecting a region and narrowing the search domain until you arrive at detailed regulatory information. In technical terms, the structure is implemented through Flask routes and templates for each page type: home.html, region.html, topic.html, category.html, and subsection.html. This page model improves scalability because each level has a defined purpose and can be extended without redesigning the rest of the interface.

4.3.2: UI Design

The user interface uses a card-based layout as its main presentation system. On the home, region, topic, and category pages, selectable content is rendered as cards arranged in a responsive grid using CSS Grid. Each card serves as both a visual grouping element and a navigation control, allowing users to quickly identify available options and navigate deeper into the hierarchy. The cards have a white background, rounded corners, and subtle shadows to visually separate them from the page background. There are also slight interactive cues, with each card shifting slightly upward when the mouse is hovered over.

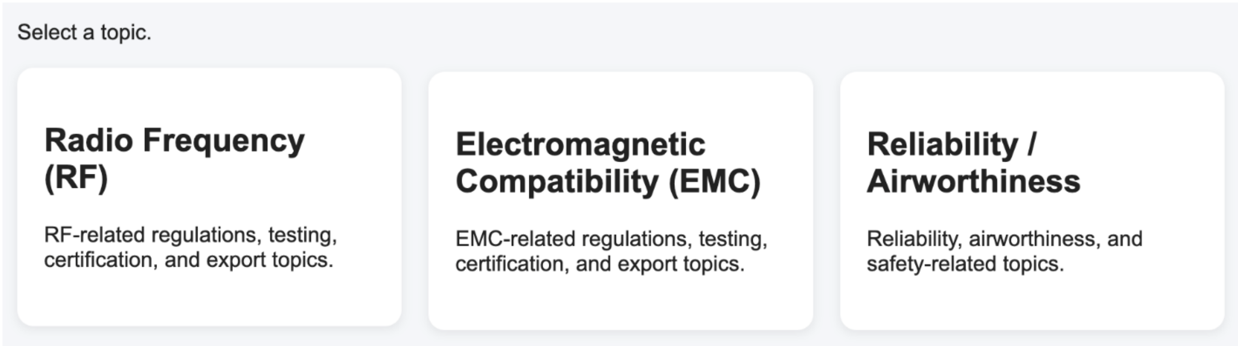

Figure 9: "Select a Topic" Page from Website

This visual design prioritizes simplicity and readability for users over visual aesthetics and eye-catching colors. The color scheme is minimal, with light, neutral background colors, dark body text, and blue links for navigation across pages. The font is straightforward with a default Arial, and the container width is constrained. This simplicity creates a clean page hierarchy, allowing headings, descriptions, cards, and backlinks to each have a distinct visual role. The interface avoids cluttering and focuses on presenting regulatory content in a more information-oriented format than media-oriented.

4.3.3: Navigation Experience

The navigation experience is intentionally straightforward. Users move forward by selecting cards and move backward through backlinks at the top of each page. This creates a predictable browsing model and reduces user disorientation when moving between multiple levels of technical content. Since each template includes a backlink, users can return to the previous page without restarting the home screen, which is useful for comparing related categories within the same topic or region.

At its current stage, the interface lacks explicit onboarding or an instruction component on the home page. While the site is still usable and the hierarchy is clear, adding a short guidance section on the landing page would improve usability for first-time users. For example, a brief sentence explaining that users first choose a region, then a topic, and then a category would make the navigation model immediately clearer. This would be a simple improvement that strengthens the interface without requiring any changes to the routing structure.

4.3.4: Front-End results

The final front-end interface successfully renders the regulatory information in a consistent and navigable format. Users start at the home page, select a region, choose a topic, move into a category, and finally view the subsection entries. The card-based layout helps separate these regions, topics, categories, and entries into readable sections. This improves usability for users, especially those unfamiliar with regulatory information. The hierarchical browsing structure works as intended, and the visual organization supports the project's goal: making complex regulatory information easy to navigate. The front end also includes a navigation bar at the top across all pages, as shown in figure 10, which provides users with constant access to the home page, keyword search, and administrator login functions from anywhere in the navigation hierarchy.

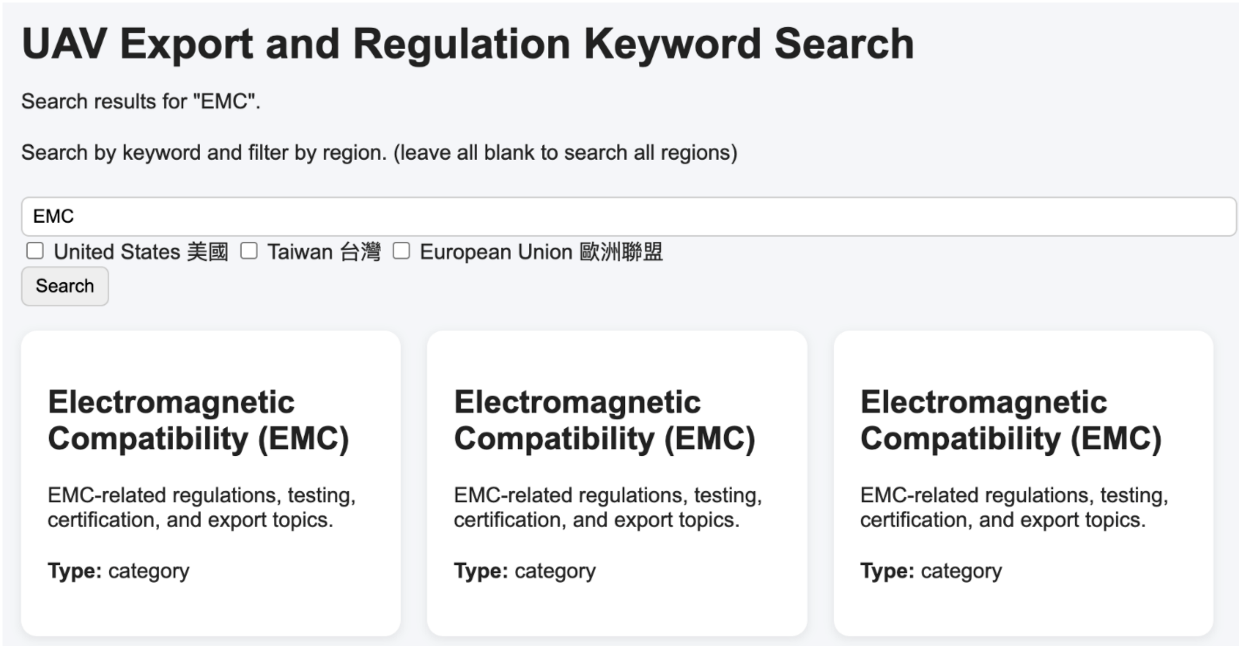

Figure 10: Website searching feature interface

The search interface is another key factor in improving front-end development and experience. Users can enter a keyword and filter by region, allowing them to locate specific information without manually clicking through the full hierarchy or blindly searching. This feature is especially useful because the website contains information across Taiwan, the United States, and the European Union. Overall, the front end is not just a demonstration layout, but a functioning user interface capable of browsing, searching, and accessing other features for accessing UAV regulatory resources.

4.4: Back End Design

The back end of the UAV Export and Regulation Interface was designed to support the website's routing, data retrieval, search functions, administrator access, and content management. While the front end determines how users view and interact with the website, the back-end controls how information is loaded, organized, protected, and delivered to each page. In this project, the back end uses Flask to connect the structured JSON data files with the HTML templates, allowing the website to dynamically generate content without requiring each page to be manually created. This structure supports scalability, maintainability, and future expansion as ETC adds new regions, topics, categories, or regulatory entries.

4.4.1: Application Framework

The website was developed using Flask, a Python web development framework that enables Python to serve web applications. Flask was selected due to its simplicity, flexibility, and minimal overhead. This allowed the team to quickly build and test core functionality without the complexity of larger frameworks. Its modular structure supports easy implementation of hierarchical navigation. This aligns with the project's design of organizing information by region and category. Python enables easy integration with data sources, such as the JSON files used in this project, while also leaving the possibility of future database integration open. Additionally, Flask includes its own development server, which can be started simply by running the Python file, and it allows for easy deployment during the development phase to ensure progress.

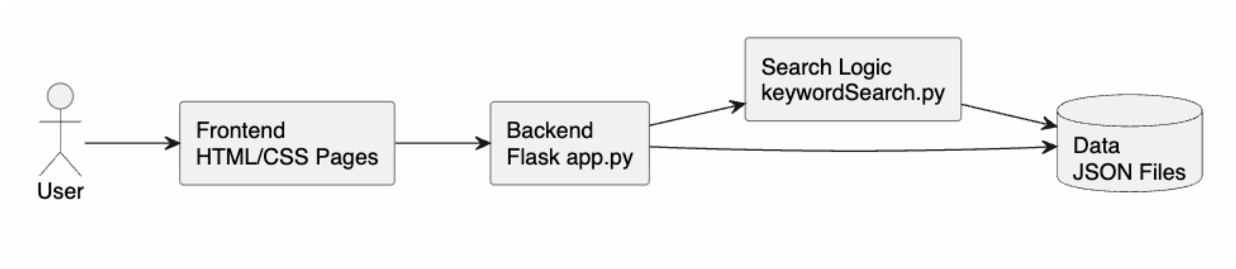

Figure 11: Website development infrastructure

4.4.2: Routing and Navigation

While the front-end section describes navigation from the user's perspective, this section explains how that navigation is implemented on the back end through Flask routes and template rendering. The routing and navigation structure of this interface was designed to guide users through regulatory information in a clear and hierarchical manner. Rather than presenting all information on one page, this system uses a layered navigation model that moves from a general home page to more specific content. The primary navigation follows this structure:

Home Page → Region Page → Topic Page → Category Page →Subsection Page

This routing design improves scalability by separating content into multiple template views, rather than concentrating all the information on a single page. This application uses dedicated HTML template pages, with each page responsible for rendering a specific content hierarchy. This modular structure supports the reuse of layouts, simplifies debugging, and makes future expansion and scalability of the interface simple.

The navigation was also made to support bidirectional movement between pages. Forward navigation through the hierarchy allows users to progressively narrow their search from generic to more specific regulatory information, while backward navigation preserves the path context and reduces user disorientation. This is especially important in an information-heavy interface, where repeated comparisons between jurisdictions may be necessary, or continuous movement between multiple broad topics. Structuring the interface with predictable interface movement improves code usability and maintainability.

At a broader architectural level, this program supports scalability and ongoing development. Because the navigation structure is template-based and data-driven, rather than hard-coded, new regulatory sections, regions, topics, or categories can be easily implemented using repeated templates without requiring a full change to the routing framework. This makes the interface suitable for iterative development, especially since the contents of this database are ever changing.

4.4.3: Data Structure

To further support scalability and maintainability, the interface stores regulatory content in structured JSON files, with one assigned to each region or jurisdiction, such as Taiwan, the United States, and the European Union. Each regional JSON file serves as a self-contained data source, organizing information by topic, category, subsection, and individual entries. This design allows the application to separate information content from presentation logic, making it easy to update content and category entries without modifying frontend templates or routing code.

At the highest level, each JSON file begins with a regional identifier and a display name. Within that region, content is then organized into major topics, such as Radio Frequency, Electromagnetic Susceptibility, or Airworthiness and Reliability. Each topic contains a title, description, and a set of categories, such as Regulation, Testing, or Certifications. This creates a uniform, identical layout and structure across all regions and topics, useful for comparative analysis and ensuring similar types of information across the board.

4.4.4: Functional Results

The completed back end successfully supports the main function of the UAV Export and Regulation Interface. The application reads regulatory content from JSON files in the data folder and dynamically generates pages from that structured content. This means that the website does not require separate hard-coded pages for every region, topic, category, or subsection. Instead, the same route and template structure can be reused as the database grows.

The keyword search function improves the database's usability. Rather than requiring users to manually browse each page, the search function can scan the content from the JSON files and return matching results from all hierarchy levels. The search system is not case-sensitive, allowing it to return matches for strings, which is useful when users don’t know the exact title, regulation, or a topic. The search results are sorted so that broader or more direct results appear first, while leaving more detailed, specific entries lower.

The back end also supports hyperlinks to external sources. Each entry lists all source titles and brief summaries and includes either an external URL or a PDF link, as applicable. This allows the website to not only function as a summary database but also as a source-access tool, where users can open original documents or reference pages in new tabs. This is important for this regulatory interface as it provides summarized guidance, but also original source material.

4.4.5: Administrative Editing Functions

A major back-end improvement was the addition of a password-protected administrator function. As shown in Figure 12, the website includes an administrator login system that uses Flask sessions to determine whether a user has editing access. When an authorized administrator is logged in, additional forms become visible on the front end, allowing them to add new regions, topics, categories, subsections, and entries directly through the website interface.

Figure 12: Admin Login Website Interface

This feature is important because the regulatory interface changes over time, enabling easy updates and new entries while restricting access for general interface users. Because the interface is publicly accessible, this administrator restriction is necessary to prevent general users from editing or adding entries. Without this access control, any visitor could potentially change the regulatory content, which would weaken the reliability of the website.

4.5: Deployment and Live Website Results

In addition to developing the interface on a local Flask environment, the final website product was deployed on Render as a publicly accessible application. Compared to the development stages, this marks a major milestone and success in the website's creation, where previously users would need to run app.py locally and access the site through a localhost address. Users would also need Python and an Integrated Development Environment (IDE) such as VS Code or PyCharm to do so. With Render deployment, the website can now be accessed directly via a web link, making it easier for ETC staff, project team members, and other stakeholders to view without the need to install Python or set up a local server.

Gunicorn was used as the production web server for the deployed Flask application. While Flask includes a built-in development server, it is primarily intended for local testing and debugging rather than public deployment. Gunicorn acts as the server process that receives web requests on Render and passes them to the Flask application. This allows the same Flask code developed locally to run in a more appropriate cloud-hosted environment. In this project, Gunicorn is important because it serves as a bridge between the Flask application and Render's hosting environment, allowing the website to remain lightweight while still being accessible through a public web link.

The deployment also demonstrates that the interface is functional outside of a development environment. The website loads through a cloud-hosted service, renders the front end, and allows users to interact with the Flask back-end hierarchy. This creates a lightweight deployment structure that fits the project's current scope while still allowing the website to function as a live public interface. Deployment via Render also improves the product's sustainability. Since ETC can access the website through a browser, the deliverable is easier to review, share, and maintain after the project ends. While future versions could use a production database, user accounts, and more advanced hosting infrastructure, the current Render deployment successfully proves that the website can function as a live information platform.

4.6: Facilities Tour

In addition to the interviews our team conducted, the ETC provided us with a tour of their facilities, giving us a better understanding of compliance, reliability, and safety testing for UAVs and other electronics. One focus of the ETC is electromagnetic compliance (EMC). EMC ensures that devices operate without causing electromagnetic interference (EMI) or being affected by unwanted electromagnetic interference (electromagnetic susceptibility - EMS). Any electronic device must pass required tests to prove it meets electromagnetic standards before it can be sold. The tours started with a presentation outlining the EMC testing categories that the ETC offers, including EMI, charging, EMS, static discharge, and UAV geofencing. We were then shown testing spaces and equipment used to evaluate how devices perform/comply while being subjected to different conditions. By receiving these tours, we were able to have a better understanding of the ETC's testing capacity, which allowed us to produce the website and add to our recommendations.

4.6.1: Static Discharge Testing

An important part of both safety and EMC testing of all electronic devices is making sure that the device will not shock or cause electrical burns to the user. The specific regulation electronic devices must adhere to is IEC 62368-1, according to which a device may not be able to transfer a current of 0.25 mA to the user after touching the device. This is the limit for devices meant for the public; however, devices meant for trained professions can have a higher standard (see 4.5.7). The test uses a human-body model electrostatic discharge (ESD) simulator that simulates the touch of human skin and measures the voltage passing through. This device must touch all the highest risk areas of the device being tested. For example, when testing drones, the ESD simulator touches all the motor's bare metal shafts. If the ESD simulator measures above the 0.25 mA threshold, then the drone does not pass, and cannot be sold to the general public.

In addition to not shocking the user, the device must also be able to withstand a static electrical discharge coming from the user. According to Taiwanese Regulation IEC 61000-4-2, the device must pass a test which simulates a human with static electricity touching the product. This test uses an ESD gun to simulate the current of a charged human body, discharging into the device using standardized voltage levels. The goal is to ensure that the drone continues to operate correctly and is not damaged when subjected to these discharges.

4.6.2: Live EMC Charge Testing

We were able to see a live demonstration of an EMC, Conducted Emissions on a commercial DJI mini 3 drone. With support from an EMT staff engineer, the charging mode of the DJI mini 3 drone was tested against IEC 61000-4-3, also known as the Radiated Susceptibility (RS) regulation. This testing was done using a field-generating antenna and a signal-reading computer. Since the DJI device being tested has already been approved for sale, the RS result was satisfactory.

The EMC staff engineer also demonstrated conducted emissions testing using the DJI device cable (charging cable). The drone was set on a table, perpendicular to the wall 30 cm away. The charging port is 80 cm away from the drone and connected by a cable. The test used a large antenna to measure any unwanted radiation that may leak while the drone is charging. The drone must not emit more than 30 dBμV/m at 10 m in the 30–230 MHz frequency, and 230–37 dBμV/m at 10 m in the 1000 MHz frequency range. Like the previous scenario, the test was passed successfully.

4.6.3: Geofence Testing

When conducting Geofencing testing, the primary device was a GPS simulator called the Global Navigation Satellite System (GNSS), which uses radio-frequency signals to replicate satellite data. This was done within an RF anechoic chamber to block out the signal from the real GPS. The GPS simulator allows the DJI drone to simulate flying at Taoyuan Airport. Based on Chapter 9-2 of the Taiwan CAA, the simulated location is deemed a NO-FLY zone; the DJI mini 3 drone's performance was tested against this regulation. This test officially verifies the geofencing application of the DJI device model. In practice, the drone should fly back to a safe area, warn the user, or stop flying once it reaches a no-fly zone. During the test, the drone's propellers are removed to prevent it from colliding with objects at the test site.

4.6.4 EMI Testing

According to regulations, only certain devices are permitted to use certain bands of the RF spectrum. For example, in Taiwan, wireless vehicle key fobs can transmit data over 125 kHz-134 kHz bands. Other devices are prohibited from transmitting within that range, and key fobs are not allowed to transmit outside that range. These regulations are essential to prevent electromagnetic interference between devices and to ensure that each device has room in the RF spectrum to transmit. Within these bands, there can be multiple channels (sub bands) that allow multiple devices of the same type to be in the same area and not interfere with each other.

These regulations apply to drones, which rely heavily on wireless communication for control, telemetry, and video transmission. Drones do not have a set bandwidth. Instead, UAV systems typically operate using standardized communication modules, such as Wi-Fi chips or proprietary RF links. The frequency on which they are allowed to transmit is determined by whichever communication device they use. For example, Wi-Fi chips are a common method of communication for drones. Wi-Fi is restricted to between 2.4 GHz and 5.8 GHz. The ETC does not run this specific type of test on the drone itself and instead will test a given Wi-Fi chip or communication device that the drone uses. During our tour we got to see this bandwidth testing performed on an ESP32. The ESP32 is widely used in both UAV and general IOT contexts.

Unfortunately, all electronic devices inherently release RF radiation on frequencies outside of their allowed range. This happens because internal electronics, such as clocks, switching circuits, and digital signals, unintentionally create extra signals. This is unavoidable, therefore, regulations dictate that these unwanted emissions must fall below a certain threshold. Specifically, the power of these emissions must be at least 20 dB lower than the desired signal within a 100 kHz bandwidth. This is tested by using the device over a period of time and measuring all of the emissions emanating from the device. As long as the threshold is not violated, the device is allowed to be used on the drone.

4.6.5: EMS Testing

It is also important to ensure that drones will not react negatively when exposed to other electronic devices emitting RF signals. EMS testing evaluates a device's ability to function correctly even in the presence of electromagnetic disturbances, such as signals from nearby wireless systems. This helps ensure the reliability and performance of the drone in real-world environments. The performance of UAV systems is tested by sending wireless signals to the drone while it is on and being controlled without propellers. Using a laser tachometer, the ETC staff monitors the speed of the motors as it receives signal interference. For the drone to pass, the speed must not change while the other signals try to interfere with its operation.

4.6.6: Reliability Testing

We also were given a tour of the labs and equipment used to perform reliability and safety testing. While reliability testing is not required by law to sell their systems, companies can require the test to determine if their product will fail under certain conditions. Often times in UAV systems, only certain critical components will undergo reliability tests, such as the battery. The main standard used is IEC 60068 for commercial drones, and MIL-STD-810 for military reliability. These regulations are guidelines for how to test drones under different conditions such as heat, cold, humidity, sudden temperature changes, vibration, or other external stresses.

This lab includes a range of testing systems. Temperature and humidity chambers control both heat and moisture levels to simulate prolonged periods outside. Thermal shock chambers test how the product handles sudden changes in temperature. Other testing machines simulate harsh, real-world conditions, such as corrosive gases, salt, dust, and low pressure (high altitude simulation). There are also solar chambers which expose devices to long periods of heat and UV radiation and mechanical tests, which assess how the system responds to physical stresses, such as vibration or drop testing. Reliability Testing can be costly, but it helps confirm that the product will perform as expected.

4.6.7: Safety Testing

The goal of safety testing is to protect the user from different types of energy. This is defined in CNS 15598-1 and follows the principles of hazard-based safety engineering. The goal is to identify a product's energy sources and reduce the risk they pose. This applies to all forms of energy, such as kinetic, electrical, and thermal energy. Risk levels are grouped into three types. Some energy is detectible but not harmful. An example of this could be shock gum (electrical) or a massage gun on a low setting (kinetic). Some can cause some pain but not serious injury, like the propellers of a drone. The highest level can cause serious harm, such as exposed high-voltage wiring that can electrocute the user. The design must prevent users from being exposed to dangerous energy.

Safety requirements also depend on the type of user. The general public has the strictest limits whereas trained users can be exposed to more risk. Products designed for experts can expose the user to the highest levels of risk. Because of this, more safeguards are required for common users and typically require that safeguards be built into the device. For example, adding propeller guards to the sides of drones to minimize injuries. The safeguards can also be provided through instructions, when devices are meant for trained users or experts. This approach ensures risks are controlled based on how the product is used and who is using it. Overall, safety design seeks to reduce exposure to dangerous energy and prevent injury during use.

4.6.8: Facilities Tours Relevance

Our facility tours provided us with a better understanding of what the ETC does on a daily basis, along with their workflow, and testing capacity. After seeing the time required to perform certain tests, and how many resources they have available, we were able to better understand where bottlenecks are in the testing process, and what could possibly be changed to improve workflow. The tours also helped us better organize the content for our website. By seeing how the tests are currently grouped at the ETC, we were able to create the website's structure in a way that reflects testing categories, such as EMC, safety, and reliability. This added to the usefulness and accuracy of the website for the user. In addition, the tours supported our recommendations. By understanding the ETC's current capabilities, we were able to recommend areas of improvement, especially over international boundaries. Without the tours, the recommendations would not reflect the ETC's real-word operations. Overall, the facility tours connected our research to its practical implementation, helped evolve our website's structure, and gave us a clear understanding of what exists and is also lacking at the ETC.

4.7: Drone Company Visits

In addition to the interviews conducted with ETC staff and tour of testing facilities, our team was also fortunate enough to visit two Taiwanese drone manufacturers, AVIX and Carbon-Based Technologies Inc. These visits gave us a better understanding of how manufacturers view certification, testing, product development, and international market access.

4.7.1: Visit with AVIX

During the visit with AVIX, their CEO presented their history, product lines, and several real-world projects. One major point was AVIX's focus on helicopter style UAVs. They explained the design benefits including higher payload capacity and superior endurance to other styles of drones. They also presented many real-world projects including work with the Taiwan Coast Guard, military-related applications, powerline inspection, emergency response, and long-range mission testing. These projects show that AVIX is well beyond prototypes and is utilizing its drones for practical applications.

One important part of the visit was understanding that a UAV is not only the aircraft, but also training and maintenance, showing that certification and testing must support the full system and not only individual components. Overall, AVIX showed that Taiwanese manufacturers are capable of developing advanced, mission-specific systems. Additionally, our visit reinforced that AVIX (and drone manufacturers in general) need clearer and more internationally aligned certification pathways. For the ETC, this is an opportunity to create a local testing regime that matches foreign certification demands. Manufacturers will benefit by having international market-ready products.

4.7.2: Visit with Carbon-Based Technologies Inc.

The visit with Carbon-Based Technologies Inc. (CBTI) gave us a different perspective on Taiwan's UAV industry. Unlike AVIX, which concentrated on helicopter-style UAVs, CBTI had a wider variety of UxV (Unmanned X Vehicles) systems. This not only included aerial systems but also marine and land systems. This was important because it showed that certification and compliance issues may apply across a greater variety of autonomous vehicles and not only UAVs. They also revealed that they have seven products certified by Taiwan's defense department, showing they are already in a strong position in Taiwan's domestic defense market. However, they said that they are currently focused on the domestic market but have no exports at present.

This made the Green and Blue UAS presentation especially important. CBTI representatives showed interest in understanding the process, costs, and testing requirements associated with future U.S. market entry. Supply chain requirements were also discussed, especially the importance of documenting the country of origin of essential components. The visit showed that CBTI already has a strong foundation in Taiwan's domestic defense market while also beginning to learn more about what may be required for future access to international markets.

4.7.3: Relevance to the ETC

The visits to AVIX and CBTI provided direct industry examples of the larger issues discussed throughout this paper. Taiwan has manufacturers with strong technical ability, but the main challenge is certification, market access, and compliance across different systems.

AVIX is focused on operations, system performance, and project experience while CBTI is at present, more focused on domestic defense work. AVIX highlighted payload, communication systems, and field projects while CBTI highlighted its seven defense-certified products and its work across air, sea, and land. These differences show that Taiwanese manufacturers need different types of support.

Both visits also showed that certification affects design, sourcing, documentation, and export planning. A company may have a capable product and still deal with delays if its certification path is unclear. This demonstrates that Taiwan has the technical foundation to compete globally, but the gap is the path from domestic capability to international acceptance.

Chapter 5: Conclusion

This project addressed a key problem in Taiwan's UAV industry: the lack of a clear, unified certification pathway for entering international markets. Taiwan has strong technical capabilities, but the certification process is fragmented and poorly aligned with its international partners. As discussed in earlier chapters, different agencies, such as the CAA, NCC, and BSMI, each handle different parts of the sales process in Taiwanese markets. While some standards are the same across international boundaries, such as the NCC largely adhering to IEC principles, this is not the case for every regulation. Compounding this problem, are frequent new regulations, such as Blue UAS certification, which complicate Taiwan's entry into international markets. As a result, Taiwanese UAV manufacturers face challenges when trying to export their products. This project intends to help the ETC align its testing services with global standards and thus simplify the process of exporting Taiwanese UAVs to the U.S. and EU.

To complete this work, our team used several research methods. The literature review helped build an understanding of U.S. and EU regulations. Interviews with ETC staff gave us insight into current testing practices and potential areas of growth for the ETC. Facility tours demonstrated how tests are performed in real-world settings. Thanks to these research methods, the team created a large dataset of regulations covering Taiwan, the United States, and the European Union. This dataset was used to support regulatory comparison and identify differences between the country's regulations. We were then able to produce a website that the ETC could host on their network, displaying the data in a user-friendly, easily modifiable way. These steps showed us that international alignment is very important.

Moving forward, this project provides the ETC with a foundation to support Taiwan's UAV industry in an international context. Becoming a Green UAS tester, similar to the organization of ITRI, is a possible solution for the ETC to pursue. This would help the ETC connect its existing strengths to newer certification needs, such as cybersecurity and supply chain trust. Additionally, the website our team created, which makes the regulation dataset easier to search, compare, and update, would be a useful tool for our sponsor. Instead of keeping all the information in a large spreadsheet, the website gives the ETC a more user-friendly platform where staff can look up different standards, add sources, and continue building their own datasets over time. These outcomes show how the ETC can play a larger role in strengthening Taiwan's UAV market and prepare it for future international opportunities.

Chapter 6: Recommendations to ETC

From the results of our thematic and comparative analyses, we developed recommendations for the ETC in navigating international regulatory frameworks. Using information on background research provided by the ETC, we developed recommendations for clear certification pathways for entering international markets. Finally with the knowledge base created during our research into U.S.-EU regulatory frameworks, we have developed tools to keep track of progress and recommendations on how to best use this information. Together, these three recommendations below cannot only provide direction for the ETC but also include a database tool to support ETC's future progress.

Recommendation 1: Navigating International Regulatory Frameworks

The ETC should continue monitoring the regulatory landscape to track international standards and those referenced in emerging regulations. Our analysis indicates that ETC aligns more closely with the U.S. regulatory system in both thematic focus and standards usage, suggesting a strong opportunity for the ETC to serve as an assessor for Blue/Green UAV certification. To support this goal, the ETC should explore partnerships with organizations such as ITRI and AUVSI.

Additionally, as countries such as Canada and India rely heavily on international standards supplemented by EU and U.S. frameworks, particularly in emergency response activities, the ETC should prioritize alignment with widely adopted international standards. This includes assessing the applicability of transferable emergency response standards from other ETC product testing. Following this, the ETC should strengthen compatibility with U.S. regulatory standards and invest in cybersecurity testing based on EU-referenced standards, along with Remote ID, electronic conspicuity, and Detect and Avoid (DAA) systems.

With the provided analysis program files, the ETC would be able to track the progression of their standards and methods as testing and certification frameworks develop in Taiwan, European Union, and the United States, and to include other countries in the future.

Recommendation 2: Applying for Blue/Green UAS Certification and Fulfilling EASA Requirements

The ETC should consider applying for Green UAS assessment opportunities through AUVSI, as this path would allow them to expand their role in UAV testing. As discussed in Chapter 2, the Blue UAS program is more stringent and for military/defense drone applications. AUVSI has created a streamlined Green UAS program by outsourcing their testing to a group of AUVSI approved assessors. Whereas it may not be realistic for the ETC to become a Blue UAS assessor at this time, Green UAS is a more practical path for the ETC as it supports commercial and non-defense UAV certification. ITRI has already shown that it is possible for a Taiwanese institution to become a Green UAS assessing body. This recommendation fits the ETC since it plays into its current strengths: EMC testing, safety and reliability, and cybersecurity-related work.

To support this recommendation, the ETC should continue developing relationships with organizations such as AUVSI and ITRI, as these partnerships could help them better understand Green UAS assessment requirements. ITRI can serve as a useful reference within Taiwan because it is already connected to the country's technology and certification environment, while AUVSI can provide a clearer pathway for understanding how Green UAS assessment works in an international context. Furthermore, the Taiwanese Ministry of Economic Affairs is a major institution and helped ITRI achieve its Green UAV status (AUVSI, 2026). Our team has also provided the ETC with a list of contacts that could make the next steps easier for ETC staff. During our lab visits, we learned that some ETC staff members were already planning a trip to Washington, DC in July to learn more about the Green/Blue UAS program. With our recommendations, they are now additionally considering extending the trip to include a visit to AUVSI's headquarters in Arlington, Virginia.

If successful, this would make the ETC the second non-American institution to become a Green UAS certifier. Information regarding the exact tests required for Green UAS certification is not publicly available; only the general evaluation categories have been disclosed. Based on the categories provided by AUVSI, a lot of the testing that has already been done at the ETC appears to match Green UAS certification, particularly with regard to NIST standards. Additional resources would need to be allocated to ensure NDAA supply chain compliance. The exact tests would be disclosed to the ETC once they are approved to become a Green UAS certifier. Our recommendations are to invest in internal capabilities to audit supply chain integrity in line with Green UAS expectations.

As a supplement, ETC should consider leveraging existing EU contacts to secure an authorized representative who can support declarations of conformity according to Regulation EU 2019/945 for Taiwanese manufacturers, essentially creating a pathway for manufacturers to sell their products in the EU market.

Recommendation 3: Continue to Update and Utilize the UAV Regulation Interface

The ETC should use the regulation website/dataset as a long-term tool for organizing, comparing, and updating UAV certification information, as international regulations will continue to change as the industry develops. Through this project, our team compiled a dataset of regulations from Taiwan, the U.S., and the EU. This was then turned into a working website that makes the information easier to search, view, and update. This provides the ETC with a clear and accessible way to track international standards over time, especially when different countries and agencies follow different certification frameworks. Instead of relying on a large spreadsheet, the website provides the ETC with a more user-friendly platform that supports internal research, regulatory comparison, and future updates.

The website can also help the ETC implement the recommendations from this project by providing ETC staff and drone manufacturers with an easier way to understand international certification requirements. Drone manufacturers can use the website to compare regulatory expectations across markets before attempting to enter international markets, while ETC staff can use it internally to support their certification regime while continuing to expand the datasets as new information becomes available. Keeping the website up to date would help the ETC strengthen its role as a UAV testing and certification center, while also giving Taiwan's drone industry a clearer pathway to prepare products for international markets.

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Appendices:

APPENDIX 1: Informed Consent Form

Informed Consent Agreement for Participation in a Research Study

Investigator: Worcester Polytechnic Institute

Contact Information: gr-taiwan26group3@wpi.edu

Title of Research Study: Technological Development and Industry Trends of Unmanned Aerial Vehicles (UAVs)

Sponsor: Taiwan Testing and Certification Center (ETC)

Introduction

You are being asked to participate in a research study. Before you agree, however, you must be fully informed about the purpose of the study, the procedures to be followed, and any benefits, risks or discomfort that you may experience as a result of your participation. This form presents information about the study so that you may make a fully informed decision regarding your participation.

Purpose of the study:

The Taiwan Testing and Certification Center (ETC) is a government-designed non-profit agency supervised by the Bureau of Standards, Metrology, and Inspection, under the Taiwanese Ministry of Economic Affairs. The ETC specializes in testing and regulation of control for imports, exports, and maintenance of products in accordance with the domestic industry.

The main purpose of this study is to present a more streamlined and efficient testing and certification process that align with global UAV standards and regulations. This process looks to provide a foundation for UAV certification in Taiwan, for high scale production of components and materials aligned with the global market, setting Taiwan up to be a strong contender.

Through this study, we will identify the current abilities of the ETC, and identify current bottlenecks and possible improvements to the certification process from industry subject matter experts, and work to present a streamlined solution process.

Procedures to be followed:

This survey should take around 20 minutes. Considering your experiences with your company, answer the following questions to the best of your ability, including yes or no questions and short answer responses.

Risks to study participants:

Please ensure that all answers you are providing through this survey do not include any data or information that is confidential to you or your company/institution. Do not risk any confidential information that is harmful to share to you or your company/institution.

Benefits to research participants and others:

Through this survey, you will be supporting the implementation of a streamlined regulation service for Taiwan's UAV production, and aligning them with global regulations for stronger market entry.

Record keeping and confidentiality:

Records of your participation in this study will be held confidential so far as permitted by law. However, the study investigators, the sponsor or its designee, and under certain circumstances, the Worcester Polytechnic Institute Internal Review Board (WPI IRB) will be able to inspect and have access to confidential data that identify you by name. Any publication or presentation of this data will not identify you.

Compensation or treatment in the event of injury:

You do not give up any of your legal rights by signing this statement.

For more information about this research or about the rights of research participants, or in case of research-related injury, contact:

- WPI Principle Investigators (Email: gr-taiwan26group3@wpi.edu_)
- IRB Manager (Ruth McKeogh, Tel. 508 8316699, Email: irb@wpi.edu)
- Human Protection Administrator (Gabriel Johnson, Tel. 508-831-4989, Email: gjohnson@wpi.edu).

Your participation in this research is voluntary. Your refusal to participate will not result in any penalty to you or any loss of benefits to which you may otherwise be entitled. You may decide to stop participating in the research at any time without penalty or loss of other benefits. The project investigators retain the right to cancel or postpone the experimental procedures at any time they see fit.

By signing below, you acknowledge that you have been informed about and consent to be a participant in the study described above. Make sure that your questions are answered to your satisfaction before signing. You are entitled to retain a copy of this consent agreement.

____________________________ Date: ____________________

Study Participant Signature

____

Study Participant Name (Please print)

_____________________________________ Date: ____________________

Signature of Person who explained this study

Additional clauses to add to Consent Agreements, as appropriate:

The treatment or procedures used in this research may involve risks to the subject (or to an embryo or fetus, if the subject is or may become pregnant), which are currently unknown or unforeseeable.

Significant new findings or information, developed during the course of the research, may alter the subject's willingness to participate in the study. Any such findings will be promptly communicated to all research participants.

Should a participant wish to withdraw from the study after it has begun, the following procedures should be followed: (list). The consequences for early withdrawal for the subject and the research are: (list).

Special Exceptions: Under certain circumstances**,** an IRB may approve a consent procedure which differs from some of the elements of informed consent set forth above. Before doing so, however, the IRB must make findings regarding the research justification for different procedures (i.e. a waiver of some of the informed consent requirements must be necessary for the research is to be "practicably carried out.") The IRB must also find that the research involves "no more than minimal risk to the subjects." Other requirements are found at 45 C.F.R. §46.116.

APPENDIX B: Interview Questionnaires

Questionnaire with Andre Pan – Drone Safety

Question	**Answer**
First, can you tell us about your position and what you do on a day-to-day basis.	Manager Laboratory Management
Who do you work more closely with? Ex. (Engineers, Executive, customers)	Boss Other Lab. Leader Customers Government & competent authority
Does the ETC provide Drone Safety testing services currently?	Yes.
What is the purpose of Drone Safety testing?	Ensure the safety of operators and the surrounding environment, including people and objects.
What are the safety standards that are used in the drone testing process currently?	CNS15598-1, which is harmonized with the international safety standard IEC 62368-1.
What safety-related tests are required for drone certification at the ETC?	Safety, EMC, Cybersecurity
What are the most critical safety risks that regulators are trying to mitigate for UAVs?	Cybersecurity & Data Hijacking
How do Taiwan's drone safety regulations compare to those in global market?	Aligns with global trends but maintains unique localized strictness, particularly inspection standards is for ICTAV equipment, not particularly for drones.
Does ETC testing provide local (Taiwan) or global safety testing?	Local testing
Are there any specific safety testing gaps posed by customers or global(non-Taiwan) markets that ETC can't fulfill yet?	Yes, we have not yet established dynamic testing equipment such as wind tunnel testing
Are there any UAV safety testing challenges that the ETC is anticipating? If so, what are they?	Assessment of drone swarms

Interview with Mike Tsai - Drone Reliability

Question	Answer
First, can you tell us about your position and what you do on a day-to-day basis.	I am a section leader in the Environmental and Reliability Testing division at ETC, a role I have held for about seven months. My work involves overseeing test planning and execution, coordinating with clients, and ensuring our procedures align with international standards.
What is the purpose of Drone Reliability testing?	Based on my background, I would expect the key factors to include environmental robustness (temperature, humidity, vibration, shock), component reliability for safety-critical parts such as motors and flight controllers, and structural integrity under repeated stress.
Does the ETC provide Drone Reliability testing services currently?	We have not yet conducted drone-specific testing. However, based on our existing capabilities, we would likely be able to offer temperature cycling, high temperature and high humidity endurance, vibration, mechanical shock, and possibly ingress protection testing.
What are the key factors that determine whether a drone is considered reliable?	It depends on the application. IEC 60068 and MIL-STD-810 are commonly used for environmental stress testing. DO-160G applies to aviation-grade components. In Taiwan, the CAA covers UAV registration and operations, though a drone-specific reliability standard is still being developed domestically.
What are the reliability tests conducted at ETC for drones?	I would expect the biggest challenge to be the lack of a single unified standard for drone reliability. Replicating realistic flight conditions in a lab — including complex vibration profiles and combined environmental stresses — also presents significant technical difficulty.
What are the specific benchmarks or standards for reliability that drones must meet?	We have not yet entered the drone testing market, but our capabilities are built on internationally recognized standards such as IEC 60068 and MIL-STD-810, which I believe gives us a reasonable foundation to compete.
What are the biggest challenges in ensuring drone reliability? Are there areas where current reliability testing could be improved?	I can only partially answer this. For FAA or EASA certification pathways, labs may need specific accreditations. Whether ETC's current scope fully covers drone product requirements

	for those markets is something we would need to investigate further.
How does the ETC reliability testing compare to those in the global (non-Taiwan) market?	While we have not yet entered the drone testing market, our capabilities are built on internationally recognized standards such as IEC 60068 and MIL-STD-810. I believe this gives us a solid foundation to be competitive, particularly in the Asia-Pacific region. The gap, if any, would likely be in drone-specific accreditation rather than in core testing infrastructure.
Are there any markets where ETC reliability testing is not accepted?	Key challenges include differing regulatory requirements across markets, ensuring our accreditations are recognized in each region, and navigating a rapidly evolving global UAV regulatory landscape. Helping clients identify which standards they need to comply with would also be part of our role.
Does ETC testing provide local or global (non-Taiwan) safety testing?	ETC is based in Taiwan, but we test internationally recognized standards, which means our reports can in principle support certification efforts in multiple markets. Whether that translates to full acceptance depends on each target market's specific requirements and whether they recognize the standards we apply.
What is the challenge when providing reliability testing for other markets or international customers? If there are examples for drones, please share them?	I would expect the main challenges to include navigating different regulatory requirements across markets, ensuring our accreditations are recognized in each target region, and keeping pace with a rapidly evolving global UAV regulatory landscape. Since ETC has not yet served international drone clients, we do not have a specific example to share at this time but helping customers understand which standards apply to their situation would likely be an important part of our role.

Interview with Eric Chen – Drone EMC & Geofencing

Eric Chen

First, can you tell us about your position and what you do on a day-to-day basis.

我工作於 ETC 的 EMC 1 部門，在公司已經 23 年，主要負責 EMC 測試、標準研究及新領域的測試技術開發，除了一般的測試之外，也須負責 ISO 品質系統的稽核工作，還有測試報告的簽署等等.

"I working in the EMC 1 Department at ETC for 23 years. i responsibilities include EMC testing, standards research, and the development of testing technologies for new products. In addition to technical testing, I am also responsible for ISO quality system for Quality Management Unit audits and check and signatory for test reports."

How does electromagnetic interference impact drone performance and reliability?

EMI 的問題會造成，無人機受到干擾，無人機會產生非意圖的動作，例如，不受控制的飛行等等

"EMI issues can cause interference in drones, leading to unintended maneuvers, such as uncontrolled flight."

What are the most common EMC-related issues seen in UAV systems?

造成 EMC 的問題有 3 個因素，干擾源、傳輸路徑及天線，無人機的干擾源為馬達、振盪器、IC、電池等等，傳輸路徑為電路板上的走線，天線為馬達跟電路板的電源及信號線、電路板間的電源及信號線、電池到電路板上的電源線

"There are three primary factors contributing to EMC issues: the interference source, the coupling path, and the antenna (radiator). In drones, the common interference sources include motors, oscillators, ICs, and batteries. The coupling paths consist of PCB traces, while the antenna effects are typically caused by power and signal lines connecting the motors, circuit boards, and batteries."

What EMC tests are required for drone certification at ETC?

依照 BSMI 及民航局規定測試有所不同，BSMI 規定測試 EMI 及 EMS，包含 RE、CE、ESD、RS、EFT、SURGE、CS、MS、DIP 等等，取決於無人機是否可以直接充電，測試項目有所不同，民航局僅規定 EMS 的 RS 測試項目，除了 EMC 之外，無論 BSMI 還是民航局，都須測試電子圍籬試驗

"Testing requirements vary between BSMI and CAA regulations. BSMI mandates both EMI and EMS tests, including RE, CE, ESD, RS, EFT, Surge, CS, MS, and DIP. The specific test items depend on whether the drone can be charged directly. In contrast, CAA regulations only specify the RS (Radiated Susceptibility) test. Beyond EMC, both BSMI and CAA require Geofencing testing."

Are there regulatory requirements related to map data usage in drones?
依據民航局所公告的圖資來使用。
"The use shall be based on the geofencing maps published by the CAA."

How do regulations address dynamic airspace changes (ex, temporary restrictions)?
依據民航局網站公告來使用
"Operation must be in accordance with the notices published on the CAA (Civil Aeronautics Administration) website."

Are there any challenges or bottle necks when performing EMC or Mapping tests? What improvements could be made to better evaluate map data or EMC systems?
必須在設計就考量 EMC 及圖資的問題，EMC 依據剛剛所提到 3 個因素來進行電路及配置設計，圖資的部份，依照民航局所公告的圖資來進行設計，圖資部份可隨時依照民航局公告內容，來進行更新
"EMC and geofencing must be integrated into the initial design phase. For EMC, circuit and layout design should address the three key factors mentioned earlier. Regarding geofencing, the design must align with CAA-published maps, which allow for real-time updates based on the latest CAA announcements."

Does the ETC testing provide local(Taiwan) or global EMC and Map Data testing?
測試需依照各國所規定的電子圍籬方法及範圍來做測試。
"Testing must be conducted in accordance with the geofencing methodologies and geographic boundaries mandated by the respective national regulations."

What is the challenge when providing EMC and Map Data testing for other markets or international customers? if there is example for drones, please share?
主要是必須了解各國所使用的法規及標準為何?並依照規定執行測試.了解測試所用的標準後，才有辦法了解測試過程所需要使用的測試設備儀器及測試方法
"It is essential to identify the specific regulations and standards of each country before executing tests. Only after understanding these standards can we determine the required test equipment, instrumentation, and methodologies for the process."

AY: Are Geofencing standards different from the EMC standards

EC: Yes they are different (It's in the presentation pg:23 that he previously gave)

AY: So the CAA is the only body that works with EMC and geofencing

EC: BSMI and CAA both require geofencing testing even if they are separate bodies

AY: Do you think that the ETC has the equipment required to perform testing for other countries

EC: Yes, however we don't know the other country's regulations which is why it's important to harmonize regulations internationally

Interview with Ariel Hsieh – Drone Certification

Question	Answer
First, can you tell us about your position and what you do on a day-to-day basis.	我的職務為業務窗口 客戶問題回覆處理及報價 案件跟催等行政相關事宜 My responsibilities include serving as the business contact window, handling and responding to client inquiries, preparing quotations, and following up on cases, as well as other related administrative tasks.
What are the key steps involved in certifying a drone through ETC?	無人機管轄機構 2 公斤以下標檢局 2-25 公斤民航局 25 公斤以上民航局 依照客戶樣品提供認証訊息,針對 ETC 所能提供的測試項目進行報價 Under 2 kg: BSMI 2~25KG: CAA Over 25KG: CAA Provide certification information based on customer samples, and issue quotations for the testing items that ETC can offer.
How does Taiwan's certification process compare to those in the EU and the United States?	台灣依重量分不同管轄機構 分標檢局及民航局,測試項目也有所不同,取得測試報告後還需要相關的文件送審核及取証

	目前在國際上沒有針對無人機的檢測產品標準 歐盟 CISPR32, CISPR35 美國 FCC Part 15b 皆為自我宣告 **In Taiwan, drone regulations are divided by weight categories.** Bureau of Standards, Metrology and Inspection (BSMI) and Civil Aeronautics Administration (CAA) share jurisdiction depending on the weight class. Testing requirements differ accordingly. After obtaining a test report, additional documentation must be submitted for review and certification. **Currently, there are no international product testing standards specifically for drones.** **EU: CISPR 32, CISPR 35** **USA: FCC Part 15b** **All of these are based on self-declaration.**
How does ETC follow or reference international certification standards?	現行無人機檢測,遵循 CISPR32, CISPR35 標準 Current drone detection follows CISPR32 and CISPR35 standards.
What are the biggest challenges in the current drone certification process? Are there gaps or limitations in Taiwan's certification system?	台灣無人機認証所遇到最大問題如下 1.單電池(包含電池組內之單電池)須符合 CNS 15364(102 版)

	電池組必須符合須符合(CNS 62133-2(107 版）第 7.3.8.1 節「振動」與 7.3.8.2 節「機械衝擊」
	2.無線射頻需要取得 NCC 認証
	
	3.重要零組件一覽表及需求零組件證書及規格書
	4.外殼需有防燃等級 UL97
	
	Here’s the English translation of the issues you listed regarding Taiwan’s drone certification:
	
	1. **Battery requirements**:
	  o Single cells (including cells within a battery pack) must comply with **CNS 15364 (2013 edition)**.
	  o Battery packs must comply with **CNS 62133-2 (2018 edition)**, specifically Section 7.3.8.1 “Vibration” and Section 7.3.8.2 “Mechanical Shock.”
	2. **Wireless frequency**:
	  o Must obtain **NCC (National Communications Commission) certification**.
	3. **Critical components**:
	  o A list of important components must be provided, along with the required component certificates and specifications.
	4. **Enclosure requirements**:
	
	The casing must meet **UL97 flame-retardant rating**.

Interview with Dr. Hsi-Hsun Yeh – Cybersecurity

Question	Answer
First, can you tell us about your position and what you do on a day-to-day basis.	I am the department head and a key manager, but I have a technical background, so I also personally handle some government projects
What is the purpose of Drone Cybersecurity testing? Who do you work more closely with? Ex. (Engineers, Executive, customers)	The main purposes of conducting drone cybersecurity testing are several: 1. To support government policy initiatives 2. To increase departmental revenue by providing cybersecurity testing services
Does the ETC provide Drone Cybersecurity testing services currently?	It depends on the situation, so there's no fized person I work closely with. For example, if it's related to testing technology, I'll contact the engineer; if it's about introducing or promoting testing services, I'll contact the client; and if it involves execution and requires support or assistance from a superior, I'll contact my supervisor
What are the standards followed for Cybersecurity testing on drones at the ETC? List them.	Yes, ETC currently has TAF accredited certification for remote-controlled drone cybersecurity testing laboratories. You can refer to the following link:
What cybersecurity-related tests are required for drone certification at ETC?	The application for TAF (Taiwan's Information Security Association)
What are the most common vulnerabilities you encounter in daily testing and certification servicing?	First, it depends on the type of device under test. If it's a cybersecurity test of a drone, one of the testing requirements in Taiwan's drone cybersecurity testing standards is that if there's any possibility of accessing sensitive data through operation, an identity verification mechanism is required, and there are also requirements regarding the complexity of the password entered for identity verification. Simply put, the vulnerability is an unauthorized operation.
How does the ETC test communication security (e.g., between drone and controller)?	The main purpose us to test whether the communication transmission between the drone and the ground control station involves the transmission of sensitive data without corresponding protection measures for such data, such as encryption, and a robust encryption mechanism is required.

Are there specific standards or regulations (FAA, FCC, CNS, etc.) that guide your testing?	The “Cybersecurity Testing Standards for Remotely Operated Drones” announced in Taiwan includes testing requirements, methods, and compliance criteria. Basically, anyone with a background in networking and cybersecurity can easily conduct the tests according to the standards.
What is the process for reporting and fixing security issues?	Basically, after the drone cybersecurity testing begins, each of the relevant test items within the testing specifications will be tested in one round to complete the initial test. If any parts do not pass,
Are there any new threats emerging in this space that need testing and certification?	Cybersecurity testing for drones differs significantly from that for other products, primarily focusing on GPS fraud and signal interference testing.
What is the challenge faced when providing cybersecurity testing for other markets or international customers? If there are examples for drones, please share them?	Not yet thought of

Interview with Dr. Roy Lin – Drone Communication

Question	Answer
First, can you tell us about your position and what you do on a day-to-day basis.	資通部 經理 處理政府之專案，研析、建置新領域之標準、檢測、驗證技術 Manager of Information and Communications Technology Service Department. Handles government projects, researching and developing standards, testing, and verification technologies for new fields.
Is there any testing done for UAV communications at the ETC currently?	目前無，但在後續的計畫當中是一大主題 Currently none, but it will be a major theme in future plans.
Are there any communication testing standards used at the ETC? List them	IEC 61850 OpenADR CNS 16014 TES-0D-02-01 CNS 15511-2 DNP 3.0 OCPP 1.6
How does the ETC follow or reference international certification standards when it comes to drone communication?	MAVLink STANAG 4586 TAICS TR-0032 v1.0 TAICS TS-0058 v1.0
How does Taiwan's certification process compare to those globally around drone communication?	台灣尚未建立無人機通訊驗證流程 Taiwan has not yet established a drone communication verification process.

What are the most critical communication risks that regulators are trying to mitigate for UAVs?	互運性議題(Interoperability)
Does the current ETC infrastructure support drone swarm testing?	No
What are the biggest challenges in the current drone communication process?	廠商各自開發自己的系統 Each manufacturer develops its own system
Does any certification apply to drone communication? And does the ETC provide testing for this certification?	STANAG 4586 ETC 無
Are there any specific communication testing gaps posed by customers or global (non Taiwan) markets that ETC can't fulfill yet?	目前的檢測缺口主要在於：驗證高階軍規級互通性標準（如北約 STANAG 4586）、在密集群飛情境下對 MAVLink 路由能力進行壓力測試。 Current testing gaps primarily lie in validating advanced military-grade interoperability standards (like STANAG 4586), conducting stress tests on MAVLink routing in dense swarm scenarios.

APPENDIX C: Other Analysis Tables

Table 2 – List of United States standards found online but not covered in ETC sourced material

cispr 11
do-362a
do-365
do-366
do-381
fcc part 88
fcc part 89
mil-std-1275
mil-std-461
mil-std-704
ul 1642
ul 3030
ul 62133-2

Table 3 - List of United States and European Union standards found online but not covered in ETC sourced materials

eu_not_covered_by_etc
do-178c
en 1155
en 12015
en 12016
en 12895
en 13241-1
en 13309
en 14010
en 16361
en 18031-1
en 18031-2
en 18031-3
en 303
en 304
en 305
en 40000
en 4709
en 4709-002
en 50065-1
en 50065-2-1
en 50065-2-2
en 50065-2-3

en 50083-2
en 50121-1
en 50121-2
en 50121-3-1
en 50121-3-2
en 50121-4
en 50121-5
en 50130-4
en 50148
en 50270
en 50293
en 50360
en 50370-1
en 50370-2
en 50385
en 50401
en 50412-2-1
en 50428
en 50470-1
en 50490
en 50491-5-1
en 50491-5-2
en 50491-5-3
en 50498
en 50512
en 50529-1
en 50529-2
en 50550
en 50557
en 50561-1
en 50566
en 55011
en 55012
en 55014-1
en 55014-2
en 55015
en 55022
en 55024
en 55035
en 55103-1
en 55103-2
en 60034-1
en 60204-31

en 60255-26
en 60669-2-1
en 60730-1
en 60730-2-14
en 60730-2-15
en 60730-2-5
en 60730-2-6
en 60730-2-7
en 60730-2-8
en 60730-2-9
en 60870-2-1
en 60945
en 60947-1
en 60947-2
en 60947-3
en 60947-4-1
en 60947-4-2
en 60947-4-3
en 60947-5-1
en 60947-5-2
en 60947-5-3
en 60947-5-6
en 60947-5-7
en 60947-5-9
en 60947-6-1
en 60947-6-2
en 60947-8
en 60974-10
en 61000-3-11
en 61000-3-12
en 61000-3-2
en 61000-3-3
en 61000-6-1
en 61000-6-2
en 61000-6-3
en 61000-6-4
en 61000-6-5
en 61008-1
en 61009-1
en 61131-2
en 61204-3
en 61326-1
en 61326-2-1

en 61326-2-2
en 61326-2-3
en 61326-2-4
en 61326-2-5
en 61439-1
en 61439-2
en 61439-3
en 61439-4
en 61439-5
en 61439-6
en 61543
en 61547
en 61557-12
en 617
en 618
en 61800-3
en 61812-1
en 619
en 620
en 62020
en 62026-1
en 62026-2
en 62026-3
en 62026-7
en 62040-2
en 62052-11
en 62052-21
en 62053-11
en 62053-21
en 62053-22
en 62053-23
en 62054-11
en 62054-21
en 62135-2
en 62310-2
en 62423
en 62586-1
en 62586-2
en 62606
en 63024
iec 15408-1
iec 27001
iec 55015

iec 60947-3
iec 60947-4-1
iec 60947-5-2
iec 60947-9-1
iec 61058-1
iec 62053-21
iec 62053-22
iec 62053-23
iec 62053-24
iso 13766-1
iso 14982
iso 22301
iso 31000

Table 4 - List of covered in ETC sourced materials but not found in United States and European extracted online materials

etc_only_not__online
ansi c63.10-2013
ansi c63.17-2013
ansi c63.4-2014
cispr 32
cispr 35
cispr32
cispr35
cns 11598-1
cns 14674-2
cns 14676-2
cns 14676-3
cns 14676-4
cns 14676-5
cns 14676-6
cns 14676-8
cns 1476-5
cns 1476-6
cns 1476-8
cns 15364
cns 15511
cns 15511-2
cns 15598-1
cns 15936
cns 16014
cns 16197
cns 62133-2

iec 17025
iec 60529
iec 61000-4-11
iec 61000-4-2
iec 61000-4-3
iec 61000-4-4
iec 61000-4-5
iec 61000-4-6
iec 61000-4-8
iec 61000-6-2
iec 610000-403
iec 61400-25
iec 61850
iec 62357
iec 62368-1
iec 62746
iec 62746-10-1
iec 62746-10-3
iso 14993
iso 17025
iso 21207
ul 2900-1
ul 94
ul97